

# Success Begets Success: The Dynamic Treatment Effects of Student Financial Aid

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## Abstract

Governments invest billions in student aid, yet the timing and targeting of aid remain open questions. In Latvia, regularly reassigned merit aid funds over 90% of all aided students. Using administrative panel data covering two public universities that enroll a third of the nation's tertiary students and a cohort GPA-rank regression discontinuity design, I estimate the dynamic treatment effects of tuition waivers and stipends. Waivers and stipends increase degree completion by 15.4 and 7.6 percentage points and improve next-semester grade point average by 0.49 and 0.26 standard deviations, respectively. I make two central contributions. First, I find initial aid crowds in future aid, showing that static evaluations substantially underestimate aid's long-term impact. Second, I show that aid operates along two distinct margins: while aid exhibits dynamic complementarity in academic effort (increasing GPAs for top performers), it is compensatory for persistence (increasing graduation for lower-performing students). Consequently, early aid acts primarily as a retention safety net, whereas late aid elicits effort. Because early success begets future aid, purely merit-based allocation increases inequality over time. These results demonstrate that financial aid design requires balancing early retention support for academically struggling students against merit-based incentives later in a student's academic progression.

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# 1 Motivation

Governments worldwide invest substantial public resources in higher education, viewing it as an engine for producing the human capital that drives modern economic growth (Deming, 2022). Many talented individuals are credit-constrained and cannot finance the optimal level of education on their own, leading to underinvestment without public support. To address these issues, countries spend vast sums on student aid; in the United States alone, this figure reached \$256 billion in the 2023-2024 academic year (Ma et al., 2024).

Despite this massive scale of investment, our understanding of aid’s long-term effectiveness is clouded by the complexity of real-world policy environments. Policymakers face a design problem: how to target and allocate aid effectively across students and over time. Should aid be primarily targeted at lower- or higher-achieving students and should it primarily be provided earlier in one’s studies or later?

Evaluating financial aid policies is hindered by a fragmented landscape of interacting federal, state, and institutional aid programs. This fragmentation creates an identification challenge where aid might crowd in or crowd out future aid. An exogenous increase in one source of aid often mechanically alter the amount of aid from other sources - a phenomenon that is typically unobserved by the economist (Denning et al., 2019; Eng & Matsudaira, 2021). Consequently, estimates from these multi-program contexts capture a net effect that conflates the behavioral response to aid with the mechanical displacement of other resources.

This paper overcomes these challenges by exploiting a setting where aid is reassigned through a regular tournament for students. Unlike the noisy multi-payer systems in other settings, the Latvian higher education system utilizes a tournament-style funding model where competitively reassigned government aid is the sole source of support for more than 90% of students. Furthermore, students follow a fixed curriculum, mitigating the risk of strategic course selection. These institutional features create an opportunity to

isolate the pure incentive effects of aid and to causally decompose the total long-term impact of an award into its components.

My analysis is guided by a conceptual framework where a policymaker allocates a financial aid budget to maximize aggregate human capital across a dual margin: student persistence (the extensive margin) and student effort (the intensive margin). In this framework, aid receipt alleviates student liquidity constraints while performance requirements elicit academic effort (Lazear & Rosen, 1981; Rosen, 1986). The technology of skill formation, however, creates a theoretical tension across these two margins (Cunha & Heckman, 2007). While self-productivity generates dynamic complementarity on the intensive margin (acting as an effort multiplier for top performers), it is ex-ante ambiguous whether aid is a dynamic complement or a substitute for persistence. If aid prevents dropouts among struggling students while top students persist regardless, optimal aid design might need to shift over time from targeting lower-performing students early in college to targeting higher-achieving students as persistence becomes inelastic. On the other hand, if aid is most effective at increasing the persistence of higher-achieving students, then aid should target high-achievers at all times.

To empirically resolve these questions, I pool administrative student-level panel data from the two largest public universities in Latvia, capturing 198,286 student-semester observations and covering 32% of all tertiary students in the country. My identification strategy exploits the institutional rule that allocates aid based on GPA rank within a cohort in a regression discontinuity design. As the probability of aid receipt changes discontinuously at the cutoff, I use this variation as an exogenous instrument for aid receipt. I then implement a dynamic RD decomposition method (Biasi et al., 2024; Cellini et al., 2010; Taylor, 2014) to recursively estimate the total long-term effect of an initial aid award, the effect of the subsequent aid it crowds in, and the marginal impact of aid, holding all future aid fixed.

I find that competitively reassigned aid has large and significant total effects on stu-

dent success. Receiving a tuition waiver impacts both the extensive and intensive margin, increasing student GPA by  $0.49\sigma$  and graduation by 15.4 percentage points (pp). Stipend receipt further impacts academic trajectories, increasing GPA by  $0.26\sigma$  and graduation rates by 7.6pp. These effects reside at the upper end of those found in the prior literature (Dynarski et al., 2023; Nguyen et al., 2019), highlighting the power of aid in a setting with strong incentives and limited outside options. Decomposing these total effects reveals a crowding-in mechanism: receiving aid in one semester significantly increases the probability of receiving it in future semesters (by up to 14.6pp for waivers and 5.1pp for stipends), which drives a substantial portion of the total long-term benefit.

Isolating the marginal impact of aid across semesters reveals that aid effects differ greatly throughout a student's degree progression. Early awards act primarily as a retention mechanism: receiving aid in the first year significantly increases degree attainment and prevents dropout, but this marginal impact diminishes in later semesters once the risk of dropout subsides. In contrast, the impact of aid on academic performance (GPA) remains strong throughout one's studies, demonstrating that later awards act almost exclusively as mechanisms that elicit academic effort rather than drive persistence.

By employing the framework of the technology of skill formation (Cunha & Heckman, 2007), I show that this dynamic arises because tournament incentives interact differently across margins of human capital accumulation. For short-term academic performance, the tournament exhibits traditional dynamic complementarity, acting as an effort multiplier for high-achieving students. However, for ultimate degree attainment, financial aid is compensatory, disproportionately retaining and graduating marginal, struggling students who would otherwise drop out. Furthermore, security alters these behavioral responses: while stipends act as a safety net for students without guaranteed funding, guaranteed waivers induce a complacency effect, where winning an additional stipend has a significantly attenuated impact on academic effort for students holding guaranteed tuition waivers.

Finally, I examine the systemic inequality consequences of this tournament structure. Because aid begets aid, purely merit-based aid tournaments compound inequality over time. The probability of aid receipt and cumulative aid amounts steadily diverge toward students from the capital and those with higher baseline high school exit exam scores. The purely merit-based targeting of aid thus rewards inframarginal students who would persist regardless.

Taken together, this paper makes three primary contributions. First, to the literature on student financial aid (Dynarski et al., 2023; Nguyen et al., 2019) and the design and targeting of financial aid (Avery & Hoxby, 2004; Bettinger et al., 2012; Bleemer & Rothstein, 2026; Burland, 2023; Durlauf, 2026; Montalbán, 2022), I provide to my knowledge the first causal estimates of a dynamic financial aid crowd-in mechanism, demonstrating that conventional static evaluations substantially underestimate the impact of financial aid, while documenting how merit tournaments inadvertently compound inequality. Second, I contribute to tournament theory (Lazear & Rosen, 1981; Rosen, 1986) by empirically measuring the option value of winning an early stage in a high-stakes, repeated competition with objective rules. Finally, by reconciling the tournament structure with the technology of human capital formation (Cunha & Heckman, 2007), I provide a dual-margin perspective to financial aid design: financial aid acts as a dynamic complement to elicit intensive effort from top performers, but functions as a compensatory safety net to ensure extensive persistence for marginal students. Thus, financial aid design is dynamic, with early aid preventing dropout from lower-achieving students and later aid increasing academic effort from higher-achieving students.

## **2 Targeting Student Financial Aid**

Consider a policymaker whose goal in providing student financial aid is to maximize total aggregate human capital. To do so, the policymaker allocates aid in a manner that pro-

vides the largest marginal return across the student population.<sup>1</sup> Financial aid operates across two distinct margins: it can increase student persistence (ensuring continued enrollment into subsequent semesters) and/or increase academic effort (enabling students to accumulate more skills during enrolled terms).

In a multi-period framework where current academic effort  $e_{it}$  accumulates into future human capital  $h_{i,t+1} = f_t(h_{it}, e_{it})$ , the total marginal contribution of an award  $A_{it}$  to expected aggregate human capital  $\mathcal{W}$  across the intensive and extensive margins is:

$$\frac{\partial \mathcal{W}}{\partial A_{it}} = \underbrace{\frac{\partial S_{it}}{\partial A_{it}} \cdot h_{i,t+1}}_{\text{Extensive (Persistence) Margin}} + \underbrace{S_{it} \cdot \frac{\partial h_{i,t+1}}{\partial A_{it}}}_{\text{Intensive (Effort) Margin}} \quad (1)$$

where, with a slight abuse of notation,  $S_{it}$  denotes both the binary enrollment indicator ( $S_{it} \in \{0, 1\}$ ) and its conditional probability  $S_{it} \equiv \Pr(S_{it} = 1 \mid h_{it}, A_{it})$ , while  $h_{i,t+1}$  reflects the human capital stock realized conditional on continued enrollment.

The policymaker allocates a fixed student aid budget across a heterogeneous cohort of students who study for  $T$  periods. Furthermore, aid can be targeted by implementing academic performance standards, which may be absolute (such as minimum GPA thresholds) or relative (such as rank-order tournaments) to elicit student effort (Lazear & Rosen, 1981; Rosen, 1986).

To model student behavioral responses, consider a forward-looking student endowed with baseline human capital  $h_t$  at time  $t$ . In each period, the student chooses academic effort  $e_t$  to maximize the expected discounted value of their academic trajectory  $V_t$ , balancing the disutility of effort  $C(e_t, A_t)$  against expected aid  $A_t \in \{0, 1\}$  and future returns, weighted by the persistence probability  $S_t(h_t, A_t)$ :

$$V_t(h_t, A_t) = \max_{e_t} \{ -C(e_t, A_t) + \beta S_t(h_t, A_t) \mathbb{E}[V_{t+1}(h_{t+1}, A_{t+1}) \mid S_t = 1] \} \quad (2)$$

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<sup>1</sup>In this way the policymaker acts as a prospective meritocrat, whose objective is to assign financial aid most effectively with respect to future human capital accumulation. In contrast, a retrospective meritocrat rewards only past behavior (Durlauf, 2026).

Conditional on persistence ( $S_t = 1$ ), human capital accumulates according to  $h_{t+1} = f_t(h_t, e_t)$ . Students are credit-constrained, which leads them to underinvest in their human capital (Becker, 1964; Lochner & Monge-Naranjo, 2012; Rosen, 1976). Receiving financial aid alleviates liquidity constraints, lowering the marginal disutility of effort and allowing students to increase equilibrium study time. Assuming self-productivity ( $\partial f_t / \partial h_t > 0$ ), baseline skills directly carry forward into future human capital. In addition, I assume dynamic complementarity in the technology of skill formation ( $\partial^2 f_t / \partial h_t \partial e_t > 0$ ), implying that higher-skilled students can more productively convert freed study time into academic achievement (Cunha & Heckman, 2007; Heckman et al., 2026).

Because students are outside compulsory education, they may drop out if continuation values fall below outside options. The probability of persistence  $p_t(h_t, A_t)$  is increasing in  $h_t$ , as higher-skilled students realize higher returns to degree completion. Similarly, persistence is increasing in  $A_t$ , as financial support lowers the net cost of enrollment. However, it is theoretically ambiguous whether the marginal impact of aid on persistence is increasing or decreasing in baseline human capital  $h_t$ . On the one hand, aid and prior skill could be complements in persistence if aid is only effective for students who possess the academic ability to clear minimum progression hurdles. On the other hand, aid and prior skill could be dynamic substitutes (compensatory) if high-ability students persist regardless of aid, meaning that the marginal persistence elasticity is highest for vulnerable students near the dropout margin.

This theoretical tension governs the optimal allocation of financial aid across margins and over time. If aid and prior skill are dynamic complements in persistence, the extensive and intensive margins align, and human capital is maximized by concentrating resources on high-achieving students across all semesters. Conversely, if aid acts compensatorily on persistence, the two margins pull in opposite directions. In early semesters when dropout risk is high, total human capital is maximized by targeting aid toward lower-performing students to secure their persistence. As students advance and persis-

tence becomes inelastic,<sup>2</sup> optimal policy shifts toward targeting higher-performing students to leverage dynamic complementarity on the intensive margin. Whether higher education policy should maintain uniform merit tournaments or dynamically re-target aid over the course of study is therefore an empirical question that hinges on whether aid and prior skill are complements or substitutes in student persistence, and the changing impact of aid as students progress through their studies.

### 3 Institutional Background and Data

The higher education system in Latvia is concentrated, with the three largest public universities enrolling half of the 76,000 total students in the country. The funding system is organized based on the dual-track tuition model, wherein some students receive tuition waivers and all other students pay full tuition. This model is common throughout post-Soviet countries, as well as in some postsocialist countries in Central and Eastern Europe (Smolentseva, 2020).

In Latvia, the primary forms of student financial aid are merit-based government stipends and tuition waivers for students.<sup>3</sup> Tuition waivers are provided to 41% of all students, and 13% of all students received stipends in 2017. In the same year, only 2.8% received financial aid from any other source. Nearly half of the students get no financial aid.

The tuition waivers are allocated to specific programs and cohorts. The government-provided financial aid for a large group of students is regularly reassigned - tuition waivers are assigned for the rest of the academic year, and stipends are assigned for one semester. All students who study in full-time programs for which the government provides waivers

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<sup>2</sup>Evidence of this decline in persistence elasticity has been documented by previous work showing that completion grants have modest impacts (Goldrick-Rab et al., 2023), whereas late-stage financial penalties can accelerate graduation (Garibaldi et al., 2011).

<sup>3</sup>Within the OECD, about half of the countries offer scholarships awarded on the basis of primarily merit (OECD, 2022).

have a chance of acquiring a tuition waiver for the next semester. Students who receive tuition waivers also have the opportunity to compete for stipends. These stipends are paid out to students as direct money transfers to their bank accounts every month for the duration of the stipend.

The government requires the aid it provides to be allocated based on merit. Merit for incoming students is defined as their performance in high-school exit exams. The students with the best grades in the exams are rank-ordered within their admitted program and receive tuition waivers. In subsequent semesters, institutions have freedom to define merit as GPA with an optional addition of other characteristics such as research output.

Tuition waivers and stipends are also extensively available, with roughly half of the bachelor's and master's students in public universities receiving tuition waivers and 9% of undergraduate students and 7% of graduate students receiving stipends. The average student in 2022 had a total monthly expenditure of €1,356, of which tuition fees for the average full-time fee-paying student comprised €313 (Koroļeva et al., 2022).<sup>4</sup>

The stipends are meant to be disbursed to the highest-achieving students receiving tuition waivers. These stipends pay €99.60 per month for the five-month duration of the semester. This stipend gets deposited into the student's bank account. The stipends cover roughly 10% of the average student's monthly expenses, excluding tuition. The student financial aid assignment mechanism is visualized in Figure 1.

[Figure 1 about here.]

Students within Latvian higher education typically follow a predetermined sequence of courses each semester, a common characteristic of many European higher education systems. This fixed curriculum means that academic competition for aid occurs over a common set of academic requirements for all students in a cohort.

I pool administrative data from the two largest public universities in the country: the University of Latvia (UL) and Riga Stradiņš University (RSU). Together, these institutions

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<sup>4</sup>The other top monthly expenditures were board (€372), room (€365), and transportation (€106).

enrolled roughly 32% of all tertiary students in Latvia in 2022. UL operates as a classical university offering a broad spectrum of programs under the Ministry of Education and Science, while RSU is a highly specialized institution focused primarily on healthcare studies under the Ministry of Health.

While their academic focus and average student demographics differ slightly, both universities operate under the exact same financial aid structure. Both employ a dual-track tuition model where competitively reassigned government aid - tuition waivers and stipends - acts as the primary, and often sole, source of financial support. While there are minor institutional variations in the exact mechanics of aid assignment (for instance, UL utilizes a strict GPA cutoff for stipends, whereas RSU incorporates secondary criteria such as undergraduate scientific output), the underlying tournament dynamics remain identical.

### **3.1 Data**

In my analysis, I use administrative student-level data from two universities: RSU and UL. This data includes semester-by-semester information on the grades, program, year, high school attended, graduation, tuition waiver, and stipend status of the students. For the RSU sample, I have data on students in all bachelor's and master's healthcare programs from 2015 to 2022, a total of 33,269 unique students that are observed each semester. For UL, I employ data from 2010 to the second half of 2020. I use data for all tuition-waiver-receiving students for a total of 63,925 student-semester observations. One challenge with the UL data is that it is only available for students who are eligible for stipends (meaning those who received waivers), and I do not observe students who are on both sides of the waiver reception margin.

However, there are differences in data availability between the two institutions. Because only RSU records data on waiver competitions, the final sample is composed of stipend competitions for both UL and RSU, and of the waiver competition for RSU. Thus,

the waiver sample is smaller than the stipend sample, and it primarily captures individuals who are enrolled in healthcare programs. Additionally, only UL stores data for student high school exit exam scores, which means that analyses that use academic achievement before college will only use data from UL.

Using student high school data, I further create two demographic variables - whether a student attended a school that provides instruction in a minority language, and whether the school is located in the capital. In Latvia, Latvian language proficiency is a significant requirement for reception of aid (waivers and stipends are only provided in Latvian-language programs), and plays an extremely important role in the job market. Regional disparities are very high in Latvia, with the capital being considerably richer and having a higher level of GDP per capita than other regions. These two variables act as proxies for students' social and economic status.

Table 1 provides the descriptive statistics for the Stipend and Waiver Marginal Samples. The stipend sample pools together data from two universities, and the waiver sample has data only for RSU. Students at the stipend margin are slightly younger on average (22.35 years versus 23.14 years) and marginally more likely to be male (0.22 versus 0.15) and originate from minority-language schools (0.14 versus 0.12). Meanwhile, the demographic characteristics of students at the waiver margin are very similar to the full waiver population, with both groups showing high rates of persistence (0.95) and similar distributions across age and geographic origin.

[Table 1 about here.]

The institutions do not store data on whether a student applied for a tuition waiver or stipend. Furthermore, RSU additionally uses a composite score of GPA and student scientific output. However, for RSU GPA is still the main factor affecting aid receipt.

The institutions also do not store data on a student's admission score. However, I observe whether a student received a waiver upon enrollment and student GPA, permitting me to evaluate the impact of aid in subsequent semesters after enrollment.

The students compete within their cohorts in a given semester, year and program - the risk set. As the programs are standardized with students taking a pre-determined list of courses in pre-determined semesters, if a student does not complete their semester, they will have to retake their semester and thus be moved to a lower cohort. The risk set is dynamic and changes every semester for a student as they might fall behind, stop out, or they might have students who fell behind or stopped out come into their risk set.

I group individuals by risk sets and then take the subset of individuals who are eligible for each type of financial aid. For stipend risk sets, these are individuals who have received tuition waivers, as it is a precondition for eligibility. For tuition waiver risk sets, these are individuals who did not receive tuition waivers upon enrollment.

As the stipends are assigned based on rank order, I create a running variable that is based on rank and recenter it around the last person to have received financial aid so that the individual who received aid with the lowest rank in their risk set is at 0. I standardize student GPA by risk set to account for more or less difficult semesters and programs.

## **4 Estimating the Impact of Financial Aid**

Separating out the direct treatment effects and dynamic treatment effects requires using both a regression discontinuity design and a dynamic regression discontinuity design. The regression discontinuity design provides me with estimates of the impact of aid that include the impact of aid crowding in future aid. The dynamic regression discontinuity design allows me to separate the impact into the marginal effect and the crowding-in effect of aid while holding future aid receipt constant.

The identification strategy exploits the institutional rule whereby eligibility for tuition waivers and stipends is determined by a student's rank based on academic performance (GPA) relative to a cutoff within a defined program-cohort-semester risk set. This rule generates discontinuities in the probability of aid eligibility at the cutoff rank, providing

quasi-experimental variation that can be used by an RD design.

To recover the estimates of the effect of various types of aid on student outcomes, I use a regression discontinuity design that compares students just below and just above the grade cutoff. The identifying assumption of my estimation strategy is that potential outcomes are continuous throughout the cutoff, i.e. that they do not discontinuously jump (Lee & Lemieux, 2010). This means that students near the rank cutoff are comparable in their potential outcomes regardless of reception of treatment, allowing me to estimate the effect of the provision of a tuition waiver or stipend on their academic outcomes.

## 4.1 Estimating the Total Impacts of Aid

I begin by estimating the total impacts of aid receipt on student outcomes. Let  $r_{ijt}$  denote the rank of student  $i$  in risk set  $j$  during semester  $t$ , and let  $Above_{ijt}$  be an indicator variable equal to 1 if  $r_{ijt}$  meets or exceeds the cutoff rank  $c_{jt}$  (i.e.,  $r_{ijt} \geq c_{jt}$ , assuming higher rank is better) and 0 otherwise. The rank  $r_{ijt}$  serves as the running variable. The central identifying assumption is the continuity of potential outcomes at the cutoff  $c_{jt}$ , conditional on the running variable (Lee & Lemieux, 2010). Under this assumption, students near the cutoff are comparable, differing primarily in their aid eligibility status induced by the discontinuity.

I use rank as the running variable as that is the one that most closely mirrors the assignment mechanism. I empirically validate this, with rank as the running variable having the strongest predictive power of aid receipt at the cutoff. The main challenge with using rank as the running variable is that it is explicitly discrete. However, using GPA has the same challenges.

I employ local linear regression to estimate the Intent-to-Treat (ITT) effect of aid eligibility on outcomes measured  $k$  semesters after the competition ( $Y_{i,t+k}$ ). This generic framework is used to estimate both immediate next-semester outcomes (such as next-semester GPA) and long-term cumulative outcomes (such as graduation). The reduced-

form model is estimated as:

$$Y_{i,t+k} = \alpha + \delta_{t,t+k}^T Above_{ijt} + f(r_{ijt}) + \theta_{jt} + \epsilon_{ijt} \quad (3)$$

where  $f(r_{ijt})$  is a linear function of rank, and  $\theta_{jt}$  represents risk set fixed effects. The parameter  $\delta_{t,t+k}^T$  captures the total effect of initial eligibility. For immediate next-semester outcomes ( $k = 1$ ), no intermediate competitions occur, meaning this total effect captures the pure marginal impact of the initial award. For long-term outcomes ( $k > 1$ ), this total effect amalgamates both the direct impact of the initial award and any indirect effects mediated through changes in subsequent aid eligibility.

While the reduced-form estimates capture the policy impact of eligibility, compliance at the threshold is imperfect. To recover the Local Average Treatment Effect (LATE) of actual financial aid receipt, I estimate a fuzzy regression discontinuity design via Two-Stage Least Squares (2SLS). I utilize cutoff eligibility ( $Above_{ijt}$ ) as an exogenous instrument for the financial aid received ( $Aid_{ijt}$ ). I then conduct additional analysis to get the per-euro impacts by replacing  $Aid_{ijt}$  with the amount of financial aid received in thousands of Euros.

The first-stage equation models the amount of aid received as a function of crossing the cutoff:

$$Aid_{ijt} = \alpha_1 + \gamma_1 Above_{ijt} + f_1(r_{ijt}) + \theta_{jt} + v_{ijt} \quad (4)$$

The fitted values from the first stage ( $\widehat{Aid}_{ijt}$ ) are then used in the second-stage equation to estimate the causal impact of aid receipt, as well as the impact of €1,000 in financial aid receipt on the respective outcome:

$$Y_{i,t+k} = \alpha_2 + \beta_1 \widehat{Aid}_{ijt} + f_2(r_{ijt}) + \theta_{jt} + \epsilon_{ijt} \quad (5)$$

For  $\beta_1$  to represent an unbiased causal estimate, the instrument must satisfy the exclu-

sion restriction, requiring that crossing the eligibility threshold impacts future academic outcomes only through the allocation of financial aid. The primary theoretical threat to this assumption in a rank-order tournament is a localized discouragement effect, or the sore loser problem. If students who narrowly miss the cutoff observe their exact marginal failure, they may experience a psychological penalty and subsequently reduce their academic effort. Such a behavioral response would violate the exclusion restriction, as the cutoff would directly impact outcomes independently of aid receipt.

However, the institutional design of the Latvian allocation mechanism mitigates this threat. The tournament is effectively blind: while students know their own GPA, they are not provided with the scores of their peers, their exact rank within the dynamic risk set, or the ex-post cutoff location. Because students cannot precisely observe how close they were to the margin, it is highly improbable that psychological discouragement jumps discontinuously precisely at the threshold, preserving the validity of the exclusion restriction.

I use data-driven bandwidth selection (Calonico et al., 2020) and robust standard errors clustered at the student level. I then perform standard RD validity tests, including density continuity tests (McCrary, 2008) and covariate balance tests to assess the plausibility of the identifying assumption.<sup>5</sup>

At UL, where stipend assignment is strictly GPA-based, compliance at the cutoff is sharp. At RSU, where aid awards depend on composite criteria, the GPA rank cutoff has a more fuzzy compliance.

There are several threats to validity to this approach. The first threat is that students could for some reason have different potential outcomes across the threshold. Although I cannot empirically test this assumption, I can build intuition on it by testing whether the

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<sup>5</sup>In baseline specifications, I select bandwidths following the procedure suggested by (Calonico et al., 2020) and cluster standard errors at the student level. I evaluate robustness to two-way clustering by both student and risk set in Appendix B and find that standard errors shift by less than 2%, leaving all conclusions unchanged. To address the discrete nature of the rank, I report confidence intervals accounting for discrete mass points (Kolesár & Rothe, 2018).

observed characteristics of students are continuous throughout the cutoff. Table 2 reports the result of the balance test for stipends and waivers respectively. What I find is that students' demographic characteristics do not vary discontinuously across the threshold.

[Table 2 about here.]

The second threat is that students could potentially self-select into receiving aid. This would lead to individuals above the cutoff being qualitatively different from those below it, meaning that they would not be comparable. In my context, this would require two things - that students choose whether to apply or not apply for aid and that they can precisely manipulate or tell what their grades and rank will be in their group.

Although students can choose to apply or not for a scholarship, this condition is necessary but not sufficient to invalidate this approach. To be able to perfectly self-select, the students would need to have perfect information on the GPA of their cohort mates and whether their cohort mates will apply for aid or not. The institution does not provide information for students on the grades of their peers and their rank in their group.

An alternative risk is that administrators could potentially selectively encourage students to apply for financial aid. For example, administrators might discriminate against individuals based on their characteristics and specifically encourage certain groups to apply. However, to do so, the administrators would have to know the student's full GPA and the GPA of their peers and be able to perfectly manipulate students near the threshold. Given the centralized nature of financial aid allocation and that students submit them electronically to the central administration and not within their departments, it is highly unlikely that administrators would be able to clearly manipulate the marginal student, as they would not know who the marginal student is. Evidence in support of this is also given by the balance test, as, if administrators selectively encouraged students to apply, I would expect discontinuities in student characteristics across the threshold.

One way to test for selection into treatment is through conducting a density test, wherein the distribution of grades just below and above the cutoff is compared (Catta-

neo et al., 2020; McCrary, 2008). Although this test is useful for continuous data, the way that the GPA is constructed in my setting does not lend itself well to the density test. The aid is assigned using semester GPA, which is the average of the GPA of courses taken that semester. The policy is that the grade for a course can only be a whole number, which creates sharp discontinuities in the densities as shown in Figure 2.

[Figure 2 about here.]

Due to the structure of the data, a regular density test would be uninformative, as it would be picking up the discontinuities in the densities due to how the data is generated. At the same time, it is also possible that there are discontinuities in the densities and that students are also selecting into treatment.

To estimate the probability of the density test failing due to the structure of the data, I use a permutation test. As the cutoff is set for each separate risk set at a specific value, I resample all of the cutoffs for the risk sets and reassign them to other risk sets. If students are selecting into treatment then the density test on the observed data should be a very extreme value. On the other hand, if the changes in densities are due to the structure of the data, then the observed values would not be extreme. I report the results of these tests in Figure 3.

[Figure 3 about here.]

Using the permutation approach, I fail to reject the null hypothesis that the observed value of the density test is different from a dataset where the cutoff is as good as random. Additionally, in section A in the appendix, I show that the distribution of grades for individual risk sets, as well as conduct a conditional density ratio test for the data to show that the conditional densities of student characteristics do not jump across the cutoff (Zimmerman, 2014).<sup>6</sup>

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<sup>6</sup>The other side of the self-selection risk is the negative selection effect wherein students who might have performed better might decrease their performance if they know the precise cutoff. The competitive nature

Figure 4 plots the first-stage jump in aid receipt probability across the cutoffs for both waivers and stipends. For waivers, being above the cutoff produces a large change in aid receipt. The stipend threshold sees a smaller jump, which is result of pooling together the sharper cutoff at UL with the more fuzzy cutoff at RSU.<sup>7</sup>

[Figure 4 about here.]

## 4.2 Estimating the Dynamic Treatment Effects of Aid

To understand the long-term consequences of aid, we must evaluate treatment effects beyond the immediate next semester. In a repeated tournament, awarding aid in semester  $t$  not only has an immediate impact, but also endogenously increases a student's probability of winning aid in subsequent semesters. Consequently, the total reduced-form effect is a combination of both the direct marginal effect of aid in semester  $t$  and all the additional aid it crowds in. To separate out these effects, I employ local linear RD methods to estimate two fundamental parameters of the dynamic system.

Let  $\delta_{t,t+k}^M$  denote the marginal effect - the average causal effect of eligibility in semester  $t$  on the outcome at  $t+k$ , holding eligibility status in all intermediate semesters constant. To formalize the decomposition of the total effect into its direct and indirect components, let  $h$  serve as an index for the intermediate periods, such that  $1 \leq h \leq k$ . The total impact operates not only through the immediate reception of aid, but also crowds in future aid, defined as:

$$\underbrace{\delta_{t,t+k}^T}_{\text{Total Effect}} = \underbrace{\delta_{t,t+k}^M}_{\text{Marginal Effect}} + \underbrace{\sum_{h=1}^k \left( \underbrace{\pi_{t,t+h}}_{\text{Effect on probability of future eligibility}} \times \underbrace{\delta_{t+h,t+k}^M}_{\text{Marginal Effect in future semesters}} \right)}_{\text{Crowding-in effect}} \quad (6)$$

of the aid reduces this risk, as students cannot precisely predict where the cutoff will be as it is based on the number of available seats, applicants, and their grades. This means that a student who might be well above the cutoff would still be unable to safely reduce their performance, as they would not be able to know beforehand where the cutoff will be.

<sup>7</sup>The results for each individual institution can be found in Appendix E

The crowding-in parameter ( $\pi_{t,t+k}$ ), which captures the average causal effect of crossing the threshold in semester  $t$  on the probability of crossing the threshold  $k$  semesters later, is estimated from:

$$Above_{i,j,t+k} = \alpha + \pi_{t,t+k} Above_{ijt} + g(r_{ijt}) + \theta_{jt} + v_{ijt} \quad (7)$$

where  $g(r_{ijt})$  is a flexible polynomial function of rank and  $\theta_{jt}$  represents risk set fixed effects.

To isolate the marginal effect ( $\delta_{t,t+k}^M$ ), I employ methods from the dynamic RD literature to recursively decompose the impact (Biasi et al., 2024; Cellini et al., 2010; Taylor, 2014). This approach isolates the direct influence pathway from  $Above_{ijt}$  to  $Y_{i,t+k}$  by mathematically removing the estimated influence flowing through intermediate eligibility changes.

This recursive estimation requires a boundary condition. Because there are no intermediate periods to crowd in aid when a student is in their last semester, the marginal impact of aid eligibility equals the total impact of aid eligibility:  $\delta_{t+k,t+k}^M = \delta_{t+k,t+k}^T$ . This fact allows me to calculate the marginal effect of aid received at a student's final semester, which then allows me to calculate marginal effects of all previous aid.<sup>8</sup>

This recursive method constitutes my primary strategy for identifying the marginal impact of aid eligibility timing on both semester-level outcomes and final outcomes, such

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<sup>8</sup>In effect, this means estimating the marginal effect for each and every future semester, so, for 3 semesters out, the estimating equation becomes:

$$\begin{aligned} \delta_{t,t+3}^M = & \delta_{t,t+3}^T - \pi_{t,t+1} \times \underbrace{\left( \delta_{t+1,t+3}^T - \pi_{t+1,t+2} [\delta_{t+2,t+3}^T - (\pi_{t+2,t+3} \times \delta_{t+3,t+3}^T)] - (\pi_{t+1,t+3} \times \delta_{t+3,t+3}^T) \right)}_{\delta_{t+1,t+3}^M} \\ & - \pi_{t,t+2} \times \underbrace{\left( \delta_{t+2,t+3}^T - (\pi_{t+2,t+3} \times \delta_{t+3,t+3}^T) \right)}_{\delta_{t+2,t+3}^M} \\ & - \pi_{t,t+3} \times \underbrace{\left( \delta_{t+3,t+3}^T \right)}_{\delta_{t+3,t+3}^M} \end{aligned}$$

as graduation. It provides insights into the effectiveness of aid provision at different stages of a student's academic progression, separate from its incentive effects on future aid receipt.

For implementation, the required  $\delta^T$  and  $\pi$  components are estimated simultaneously via a system of equations using stacked regressions to obtain the full variance-covariance matrix of the component estimates. The marginal effects ( $\delta_{t,t+k}^M$ ) are then computed recursively using Equation 6. Standard errors for the baseline recursive decomposition are calculated using a block bootstrap and, as an additional robustness test, using the Delta method.

To recursively isolate the marginal effect  $\delta_{t,t+k}^M$ , the estimator relies on a combination of several total effects estimators ( $\delta^T$  and  $\pi$ ) that are generated across different semesters, which requires three identifying assumptions. The first assumption is that for each  $\delta^T$  and  $\pi$  estimand, the potential outcomes must be continuous at their respective cutoff. Second, because the eligibility cutoff changes dynamically from semester to semester, altering the sample composition of the marginal population, I assume that the local complier populations are sufficiently stable such that these dynamic parameters are homogenous across these changing risk sets. Third, the recursive estimator assumes additive separability - that the marginal impact of future aid is independent of a student's past aid history.

The institutional features support the first two assumptions. The first assumption of continuity of potential outcomes is similar to the continuity of potential outcomes discussed in the previous section and is supported by the same arguments. Regarding the second assumption, the primary threat to complier stability is that fluctuating cutoffs might capture systematically different types of marginal students over time. However, because students are locked into a fixed, pre-determined curriculum within their specific cohort, they are unable to strategically switch risk sets or alter course difficulty to game the shifting cutoffs. Consequently, the underlying complier pool remains structurally stable across periods.

The third assumption of additive separability requires that the marginal return to aid in a given semester does not depend on a student’s past aid history. I test this assumption in Section 5.3 by estimating a retrospective regression that interacts current aid assignment with past aid receipt ( $Aid_{it} \times PastAid_{i,t-1}$ ). Across most specifications, this interaction is small and statistically indistinguishable from zero on effort and persistence, with a marginally negative interaction on graduation for cash stipends. If past aid diminishes the marginal return of future aid, then using average future returns overstates the indirect crowd-in effect. Subtracting this overstated crowd-in means that the recursive estimator yields a conservative lower bound for the true direct marginal effect of early aid.<sup>9</sup>

## 5 Results: Total and Dynamic Impacts of Aid

### 5.1 The Total Impact of Financial Aid in a Competitive System

This section presents the overall impact of receiving financial aid (tuition waivers or stipends) on student effort and persistence throughout their studies, as estimated by a standard regression discontinuity design (Equation 3). These total effects ( $\delta^T$ ) reflect the consequences of awarding aid within the existing institutional framework, where aid is competitively reassigned each semester.

These estimates correspond to the impact of aid eligibility in a system where aid is competitively reassigned and answer the question of “what is the impact of being eligible for financial aid, given that it will be reassigned later?”. These effects include both the marginal effects of aid eligibility, such as increased performance in the semester of aid reception, and the indirect effects of aid eligibility, such as increased probability of aid eligibility in future semesters, which in turn increases performance in future semesters, and so on.

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<sup>9</sup>Additionally, I conduct Monte Carlo simulations in Appendix D showing that across the empirical range of plausible interaction values, the recursive estimates vary by less than 3%.

### 5.1.1 Impact of Waiver Receipt

In this subsection, I analyze the effects of acquiring a waiver irrespective of previous and future history of reception of aid. This corresponds to the estimation of the treatment effects in equation 3 and provides estimates on the effect of acquiring a waiver for the next semester when the counterfactual is not acquiring it for the next semester. This is the margin under which many scholarships operate, as students have many opportunities for the acquisition of aid throughout their studies.

I now turn to estimating the impact of the reception of waivers. I first plot the impact of the provision of waivers in Figure 5, and report the results in Table 3.

[Figure 5 about here.]

[Table 3 about here.]

I first estimate the Intent-to-Treat (ITT) effect of waiver eligibility, which captures the policy-relevant impact of crossing the threshold. I find that simply being eligible for a waiver significantly alters a student's academic trajectory. Specifically, waiver eligibility increases standardized next-semester GPA by  $0.378\sigma$ , boosts semester-to-semester persistence by 10.2 percentage points, increases the probability of being on track<sup>10</sup> by 8.9pp, raises the ultimate graduation rate by 12.4 percentage points, and decreases the probability of dropping out by 11.5 percentage points.

To get at the treatment-on-the-treated estimates, I use 2SLS to instrument for waiver receipt. I find that receiving a tuition waiver increases the student's next-semester persistence by 13.3pp, probability of remaining on track by 11.5pp, and graduation by 15.4pp.

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<sup>10</sup>Being on track is defined as the student's semester number being the next number after their current semester. In Latvia, as in many countries in Europe, students choose to study in specific programs upon enrolment, which have a pre-defined course and semester sequence.

### 5.1.2 The Impact of Stipend Receipt

Figure 6 provides a graphical representation of the changes in student outcomes across the threshold for the combined sample. The plots show the changes in outcomes based on the student's rank distance from the marginal peer who received aid in their respective risk set.

[Figure 6 about here.]

Table 4 reports the regression estimates of the impact of aid receipt on academic outcomes. By instrumenting actual stipend receipt with cutoff eligibility, I find that stipends have a powerful effect on student effort and academic progression.

[Table 4 about here.]

I find that receiving a stipend yields substantial academic returns across both the intensive and extensive margins. Specifically, the 2SLS estimates show that stipend receipt increases a student's next-semester GPA by  $0.257\sigma$ . Stipends improve academic progression: they increase semester-to-semester persistence by 5.3 percentage points, the probability of remaining on track by 7.2 percentage points, and the ultimate graduation rate by 7.6 percentage points (with corresponding ITT effects of 4.5, 6.1, and 6.6 percentage points). Scaling these estimates per €1,000 of received aid indicates that even small liquid cash transfers can have substantial impact when allocated via tournaments.

While these pooled estimates capture the generalized mechanism of the tournament, the baseline effects do vary slightly depending on the specific institution. How these effects differ by institution is detailed in Appendix E.

## 5.2 Decomposing Dynamic Treatment Effects: Marginal Impacts, Crowd-In, and Optimal Timing

The previous section established the total impact of financial aid in a system where aid is reallocated based on ongoing performance. However, these total effects amalgamate the immediate, direct impact of receiving aid in a given semester with the indirect impact that arises because aid receipt can improve performance and thus increase the probability of receiving aid in subsequent semesters (a crowding-in effect). This section aims to disentangle these components using the dynamic regression discontinuity framework.

Specifically, I first examine the extent to which current aid receipt affects future aid probability ( $\pi_{t,t+h}$ ). I then isolate the marginal impact of a single aid instance ( $\delta_{t,t+k}^M$ ), holding future aid receipt constant. Finally, I investigate how the timing of aid (early vs. late in a student's academic progression) affects long-term outcomes like graduation.

### 5.2.1 Aid Crowd-In over Time

I conduct further analysis to evaluate the effects of being offered aid on student outcomes in subsequent semesters. I use lagged values of the dependent variables to explore the impact of the provision of aid in future semesters and report them in Table 5

[Table 5 about here.]

I generally find that being eligible for either type of aid has considerable crowding-in effects on future aid. Being eligible for a waiver increases their probability of receiving a waiver for up to five semesters ahead, increasing their probability of receiving a waiver in five semesters by 14.6pp, and crowding in additional aid amounts. This effect is driven by individuals persisting through their studies, as being eligible for waiver increases persistence by 6.2pp after five semesters.

Similarly, stipend receipt also crowds in additional stipends, with individuals after five semesters being 5.1pp more likely to hold a stipend. This effect, however, seems to

be driven primarily by student effort, as stipend receipt increases future student GPA.

### 5.2.2 Intertemporal Impact of Aid: Early vs. Late Aid on Graduation

In this subsection, I explore whether the impact of aid eligibility on long-term success, specifically graduation, differs depending on when it is received in a student's academic trajectory (e.g. early vs. late). I estimate the marginal impact of aid for different semesters.

To estimate the impact of aid eligibility in any given semester, I first subset the data to include students who were undergraduates and who had a program length of 8 semesters (the most popular program length for undergraduate study), and who started and whose expected graduation falls within the timeframe of the data.

The goal of this subsetting is to ensure that the estimates are from individuals who are observed throughout the time period that I have data for and who have the same program length, to ensure that effects are not driven by attrition at the program level. In this section I report the estimates for waivers due to their impact, the results for stipends are available in the appendix. I report my estimates in Figure 7

[Figure 7 about here.]

I find that the impact of waiver eligibility is higher in earlier semesters for most, with the effect only being statistically significant in the first year or first two semesters for increasing graduation and decreasing dropout.

In Figure 8 I plot the probability of ever dropping out by individuals in each semester. This plots the probability of ever dropping out of the program for all individuals who have been observed in that semester. What I find is that the probability of dropping out is highest in the first semesters, after which it decreases, only jumping again at the final semester.

[Figure 8 about here.]

Additionally, I take the subset of individuals who were studying in 6-semester-long undergraduate programs (the most popular choice among students in the stipend sample), and conduct a decomposition into the marginal and indirect effects (Figure 9). I find a similar pattern with the impact of aid on student persistence being highest in the earlier semesters, with the effect decreasing the second half of one's studies. At the same time, however, the impact on cumulative GPA stays strong throughout one's studies.

[Figure 9 about here.]

### 5.2.3 Marginal Effect of Aid vs Crowding-In

I now turn to decomposing the total effect in every semester into the marginal effect (driven by aid eligibility in a given time) and the indirect effect (driven by future aid eligibility). I thus decompose the results for waivers in Figure 7 into the total, marginal, and crowding in effects in Figure 10 based on equation 6.

[Figure 10 about here.]

Decomposing the effects into the marginal and crowd-in for stipends (Figure 11), I find that the marginal effect plays a larger role compared to the crowd-in effect for earlier semesters for stipends. However, the crowd-in effect is most pronounced in the earliest semesters and decreasing over time.

[Figure 11 about here.]

To quantify the dynamic crowding-in effects of financial aid, I measure the crowd-in multiplier, which is the ratio of the total reduced-form effect ( $\delta_{t,t+k}^T$ ) to the direct marginal effect ( $\delta_{t,t+k}^M$ ).<sup>11</sup> If there is no crowding-in or crowding-out of aid, then the two effects are identical and the multiplier equals 1. I report the crowd-in multipliers across the first three semesters for my dynamic RD decomposition analysis in Table 6.

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<sup>11</sup>In the framework of multi-stage elimination tournaments (Lazear & Rosen, 1981; Rosen, 1986), this crowd-in captures the continuation option value of remaining in the tournament, which sustains effort and competition in intermediate rounds.

[Table 6 about here.]

For stipends in Semester 1, the multiplier is 1.45 for GPA, 1.30 for on-time graduation, and 1.25 for overall degree completion, where crowded-in aid drives 19.7% to 31.1% of the initial effects. In Semester 2, the stipend multiplier reaches 1.77 for cumulative GPA and 1.52 for on-time graduation, with future crowded-in aid accounting for 43.5% and 34.4% of the total reduced-form impacts, respectively. For tuition waivers, early point estimates similarly indicate sizable dynamic multipliers in Semester 1 ( $M = 1.90$ , share 47.3% for graduation). For later semesters, the effects attenuate as both the marginal and total effects become smaller.

### 5.3 Dynamic Complementarities in Financial Aid

The unique mechanism of aid allocation also allows further exploration of treatment effect heterogeneity of financial aid based on students' previous stock of skills and aid history. Based on the framework by Cunha and Heckman (2007), individual investments in human capital are more effective for individuals with higher a priori levels of human capital, as they are able to produce additional human capital more efficiently, creating dynamic complementarity in investment. Additionally, the sequential nature of tournaments also allows me to evaluate heterogeneity of effects by aid history.

To explore dynamic complementarity, I assume past cumulative GPA as a proxy for human capital and use a two-stage least squares approach, wherein I estimate the following equation:

$$GPA_{i,t+1,j} = \beta_0 + \beta_1 CGPA_{itj} + \beta_2 Aid_{itj} + \beta_3 (CGPA_{itj} \times Aid_{itj}) + f(r_{itj}) + \theta_{tj} + \epsilon_{it} \quad (8)$$

Which is equation 3 with an additional interaction of aid and previous CGPA. And I

instrument  $Aid_{itj}$  and  $(CGPA_{itj} \times Aid_{itj})$  using the following equations:

$$Aid_{itj} = \pi_{10} + \pi_{11}Above_{itj} + \pi_{12}(CGPA_{itj} \times Above_{itj}) + \pi_{13}CGPA_{itj} + f(r_{itj}) + \theta_{tj} + v_{1it} \quad (9)$$

$$(CGPA_{itj} \times Aid_{itj}) = \pi_{20} + \pi_{21}Above_{itj} + \pi_{22}(CGPA_{itj} \times Above_{itj}) + \pi_{23}CGPA_{itj} + f(r_{itj}) + \theta_{tj} + v_{2it} \quad (10)$$

This approach allows me to use the plausibly exogenous variation caused by the jump in aid receipt probability upon crossing the threshold to partial out the impact of aid specifically for individuals with different previous levels of human capital. The coefficient on  $\beta_3$  in the second stage equation would capture the effects of dynamic complementarity in skills and provide evidence on whether the investments in the skills of individuals are more effective for individuals with higher skills a prior. I report the results in Table 7 for waivers and stipends on next-semester GPA.

[Table 7 about here.]

[Table 8 about here.]

Contrary to a pure dynamic complementarity framework, I find mixed and margin-specific results. The interaction between prior human capital and financial aid depends fundamentally on whether we examine short-term academic effort (the intensive margin) or ultimate degree attainment (the extensive margin). For short-term academic effort, there is suggestive evidence of dynamic complementarity.

Table 7 shows that the interaction between a student's current CGPA and stipend receipt on next-semester GPA is positive ( $0.061\sigma$ ). This indicates that for already high-achieving students, winning a tournament prize can act as an effort multiplier.

When evaluating long-term extensive margins, however, aid acts in a compensatory manner. Table 8 demonstrates that the interaction between aid and prior human capital

(current CGPA) is strongly negative for degree attainment. The interaction of stipend receipt with current CGPA on graduation is -0.040, and with lagged past aid is -0.024. Waivers exhibit an even stronger compensatory interaction with current CGPA (-0.136).<sup>12</sup>

Rather than simply compounding the success of the already-resourced, financial aid is also effective at helping marginal, struggling students cross the finish line. Thus, the tournaments successfully operate across two margins - increasing the academic effort of high-achieving students and also providing support for persistence for lower-achieving students.

Furthermore, these estimates provide empirical evidence in support of the assumption of additive separability in the dynamic RD design, as the interaction between current and lagged aid is statistically indistinguishable from zero across almost all specifications, with only a marginally significant negative interaction for graduation in stipends. Additionally, these estimates allow me to formally sign the bias. Because the interaction of aid with prior human capital for cumulative GPA is positive, the average future marginal effects ( $\delta^M$ ) used in the decomposition would understate the true indirect benefit for the specific students who were crowded into receiving future aid. Thus, the decomposition would subtract a calculated crowd-in sum that is too small, leaving the estimated marginal effect of early aid on GPA as an upper bound. Conversely, for graduation, the estimated marginal effect represents a conservative lower bound. Because the interaction on past aid and prior human capital is negative on the extensive margin, the average future effects overstate the true indirect benefit for already-treated students. By subtracting this overstated indirect effect, the resulting isolated marginal effect on graduation is biased downward and represents a lower bound. Additionally, to quantify the empirical impact of this potential bias, I run a Monte Carlo simulation (Appendix D) and find that violations of additive separability between  $-0.06$  and  $+0.06$  (the empirical bounds in Tables 7

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<sup>12</sup>These compensatory effects on extensive margins are particularly pronounced in high-stakes environments without guaranteed safety nets, such as is the case in UL. A detailed breakdown of dynamic complementarity by specific institution is provided in Appendix E.

and 8) alter the recursive estimates by at most 3%.

## 5.4 Designing the Tournament

I next evaluate how the structure of the tournament - award inclusiveness, commitment horizon, and safety nets - shape student responses. Theoretical models of credit constraint predict that marginal returns to financial aid should be largest among students facing higher opportunity costs or with lower baseline levels of academic achievement.

The sequenced nature of aid, where the cutoff percentile changes from semester to semester, creates additional opportunities to investigate heterogeneities based on various financial aid program design characteristics and the characteristics of recipients. To explore these questions, I run further analysis by demographics, adding interaction terms with being above the cutoff in my regressions with various student demographics and characteristics of the provision of aid itself, such as its timing and the percentile at which the cutoff occurred. Table 9 reports the estimates for the heterogeneity of effects by student demographics.

[Table 9 about here.]

What I find is that generally the aid effectiveness does not vary by student demographics, with some slightly larger impacts on persistence and progression for students from the capital, but only for stipends.

The way a competition is designed can both help and hurt student performance. On the positive side, tournaments motivate students in two main ways. First, winning increases the chance of winning again in the future, giving students a long-term reason to try hard. Second, these tournaments are especially good at pushing the highest-performing students to work even harder. However, there is a downside. If students feel they have no chance of winning - for instance, if awards are only given to a tiny percentage of participants - they can become discouraged and perform worse.

I estimate the impacts of financial aid by the characteristics of the risk set, at which percentile within the risk set the cutoff occurs, as well as how late throughout one's studies. Table 10 reports the results.

[Table 10 about here.]

I explore the option value of financial aid as described by the literature on performance tournaments by looking at the impact of financial aid at the end of one's studies when there is no future option for renewal. I find that there are considerably stronger effects on persistence for students in the first half<sup>13</sup> of their studies, with students in the first half receiving stipends having increased probability of persistence by 2.5pp, but lower impact on GPA by  $0.14\sigma$ . Similarly, students receiving waivers in the last year have smaller impacts on persistence and probability of being on track, and students receiving waivers in their last semesters seeing smaller impacts on their dropout and graduation rates.

Exploring heterogeneity by cutoff characteristics, I find evidence that financial aid is more effective at lower percentiles. For waivers, moving from the bottom to the top percentile of the cutoff decreases the impact on persistence by 30.8 percentage points and the probability of being on track by 31.2 percentage points. This implies that financial aid tournaments are more effective when the number of awards relative to the number of participants is larger, which would align with the theoretical predictions of individuals being incentivized by tournaments more if they believe that they have higher chances of winning.

I also explore the risk versus security effects of financial aid. The impact of the tournament can also vary by the intensity of competition. To estimate the impact of increased levels of competition and risk, I leverage several sources of variation.

First, I leverage the fact that waivers, if provided starting in the fall semester, are provided for the whole academic year or two semesters. Meaning that individuals who

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<sup>13</sup>The reason that the first half is used rather than semesters is due to the fact that the university has very different program lengths, from 6-year programs for medical professionals to 2-year master's degrees

receive financial aid in the fall should feel less of a competitive effect in the first semester, as they have secured financial aid for the next semester irrespective of their performance.

Second, I explore heterogeneity by the level of difficulty of the risk set. To do so, I use two different measures of difficulty: average GPA and the historical progression leave-one-out rate for the cohort and semester. Finally, I also leverage the fact that individuals who begin their studies with a tuition waiver have the waiver guaranteed for the rest of their studies. Thus, they are less at risk of losing both types of financial aid. I explore heterogeneity for the impact of stipends for this group. I report the results in Table 11.

[Table 11 about here.]

Generally, financial aid appears to be most effective for individuals who are struggling academically, with security effects increasing persistence, but decreasing student effort. Waivers exhibit compensatory effects in higher-difficulty cohorts: below-median GPA risk sets see a  $0.35\sigma$  larger impact on next-semester GPA and an 8.9pp increase in persistence. Similarly, in below-median progression cohorts, waivers generate substantial extensive-margin gains, increasing persistence by 11.3pp and on-track progression by 10.0pp (Table 11). Stipends reinforce this compensatory pattern on the extensive margin: in below-median progression cohorts, stipends increase persistence by 5.7pp, on-track probability by 8.6pp, and ultimate graduation by 6.9pp ( $p < 0.001$ ). Furthermore, security significantly alters the behavioral response to stipends. For students with the security of a guaranteed tuition waiver, this safety net actually induces a complacency effect, with decreased impacts on academic effort (GPA) by  $0.29\sigma$ .

I also explore institution-specific heterogeneity in Appendix E. I find a consistent dynamic of aid across both institutions: early stipends act primarily as a retention mechanism to reduce dropout, with diminishing effects later on. Furthermore, in settings where many competitors already possess the safety net of a guaranteed tuition waiver (such as RSU), the marginal impact of a stipend is heavily concentrated on increasing student ef-

fort (as measured by GPA), whereas in higher-stakes environments without guaranteed waivers (such as UL), stipends drive increases in ultimate persistence and graduation.

## 5.5 Distributional Consequences: Aid Accumulation and Inequality

Although I do not find heterogeneity in the impact of student financial aid for students at the cutoff, it is important to note that the students at the cutoff are already a selected group. I thus turn to studying the impact of purely merit-based financial aid within cohorts and the impacts on inequality. To do so, I run an event study with program and semester fixed effects to compare how the probability of aid receipt evolves over the course of one's studies by finishing high school in the capital (Figure 12), HS exit exam scores (Figure 13), and gender (Figure 14).

[Figure 12 about here.]

[Figure 13 about here.]

[Figure 14 about here.]

Across both waivers and stipends, the financial aid is compounding inequality, with the aid being more likely to go to individuals who completed high school in the capital and individuals who had better high school exit exam scores. Stipends, which are more competitive, are also larger contributors to inequality, with considerably larger gender-based divergences and greater prevalence among those from the capital.

## 6 Discussion and Conclusion

In this paper, I estimate the effect of financial aid that is competitively reassigned each semester based on student merit. Using a unique institutional setting in Latvia where tuition waivers and stipends are the only form of aid available to the vast majority of

students, I provide causal evidence on the dynamic effects of these programs on student success.

I find that both tuition waivers and stipends have a significant positive impact on student outcomes, increasing their grades, persistence, and graduation rates. The total effects are large: receiving a tuition waiver increases a student's probability of graduation by 14 percentage points and their persistence into the next semester by 13 percentage points.

These estimated impacts are at the upper end of the range reported in prior financial aid evaluations, which is likely driven by the tournament-style incentives and the lack of alternative aid sources in this setting. The robustness of these findings is enhanced by the fixed curriculum, which limits students' ability to game the system through course selection and provides a cleaner measure of aid's impact on academic effort.

The central contribution of this paper is the identification of a powerful and persistent crowding-in effect, where receiving aid in one semester significantly increases the probability of receiving it in subsequent semesters. This crowd-in also contributes to increasing inequality rather than ameliorates it, as aid accumulates for individuals who have completed high school in the capital and have strong high school exit exam results.

I decompose the total effect of aid into its direct marginal impact and its indirect crowding-in impact. This analysis reveals that the crowding-in of future aid is a substantial driver of the large long-term benefits, particularly for aid awarded to students earlier in their studies.

While the baseline recursive decomposition model relies on an assumption of additive separability for tractability, I explicitly test the boundaries of this assumption by exploring dynamic complementarities in skill formation. By identifying that aid acts as a dynamic complement for short-term effort but as a compensatory substitute for long-term degree attainment, I am able to formally sign the bias of the dynamic estimator. I find that the marginal effects of early aid calculated in the decomposition represent an upper bound

for academic performance, but a lower bound for graduation.

Diving deeper into the mechanisms of skill formation, this paper evidence of the interaction between existing skills and financial aid investment. For short-term academic performance, stipends can act as a complement, multiplying the effort of students with higher baseline GPAs. However, for ultimate degree attainment, financial aid is compensatory. The impact of both waivers and stipends on graduation is significantly magnified for students with lower existing levels of human capital and fewer prior resources. This presents a nuanced consideration for policymakers: while tournament structures effectively incentivize the top of the distribution to maximize effort, a large part of the value of merit aid lies in retaining and graduating students at the bottom of the distribution who would otherwise drop out.

The results highlight that the optimal targeting of aid evolves over the students' degree progression. Early in their studies, when dropout risk is high, financial aid is most effective when targeted at lower-achieving individuals to ensure their persistence. In later semesters once students are securely on track to graduate, aid is more effective when targeted at higher-achieving students to elicit additional effort.

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Table 1: Summary Statistics for Combined Sample

| Variable      | Stipends - Full |          |        | Stipends - Marginal |          |       | Waivers - Full |          |      | Waivers - Marginal |          |      |
|---------------|-----------------|----------|--------|---------------------|----------|-------|----------------|----------|------|--------------------|----------|------|
|               | Mean            | Std.Dev. | N      | Mean                | Std.Dev. | N     | Mean           | Std.Dev. | N    | Mean               | Std.Dev. | N    |
| GPA           | 0.00            | 0.97     | 161059 | 0.32                | 0.86     | 39979 | -0.28          | 0.98     | 3024 | -0.25              | 0.95     | 1687 |
| Persisted     | 0.71            | 0.45     | 198286 | 0.76                | 0.43     | 39979 | 0.95           | 0.21     | 3024 | 0.95               | 0.21     | 1687 |
| On Track      | 0.66            | 0.47     | 197629 | 0.65                | 0.48     | 39862 | 0.96           | 0.21     | 3023 | 0.95               | 0.21     | 1686 |
| Graduation    | 0.60            | 0.49     | 156412 | 0.68                | 0.47     | 35953 | 0.80           | 0.40     | 1928 | 0.80               | 0.40     | 1142 |
| Graduation OT | 0.45            | 0.50     | 155632 | 0.52                | 0.50     | 35920 | 0.69           | 0.46     | 1928 | 0.68               | 0.47     | 1142 |
| Aid           | 0.13            | 0.34     | 198286 | 0.34                | 0.48     | 39979 | 0.40           | 0.49     | 3024 | 0.44               | 0.50     | 1687 |
| Male          | 0.15            | 0.35     | 194979 | 0.22                | 0.41     | 39243 | 0.15           | 0.36     | 3024 | 0.13               | 0.34     | 1687 |
| Age           | 23.14           | 6.17     | 177231 | 22.35               | 4.98     | 38895 | 21.78          | 5.90     | 2969 | 21.93              | 5.89     | 1659 |
| Minority      | 0.12            | 0.32     | 174821 | 0.14                | 0.34     | 38130 | 0.16           | 0.36     | 2969 | 0.15               | 0.35     | 1659 |
| Capital       | 0.40            | 0.49     | 174821 | 0.43                | 0.50     | 38130 | 0.41           | 0.49     | 2969 | 0.40               | 0.49     | 1659 |
| Math          | 71.64           | 18.18    | 23178  | 73.26               | 18.11    | 10134 | -              | -        | -    | -                  | -        | -    |

*Note:* The table reports summary statistics for individual-semester dyads. Math represents the high school exit exam percentage and is only available for the UL sub-sample.

Table 2: Balance Test

|             | Stipends         | Waivers           |
|-------------|------------------|-------------------|
| Male        | 0.003<br>(0.007) | −0.017<br>(0.031) |
| Age         | 0.137<br>(0.085) | 0.310<br>(0.408)  |
| Capital     | 0.014<br>(0.010) | −0.063<br>(0.047) |
| Minority    | 0.005<br>(0.006) | 0.017<br>(0.034)  |
| Risk Set FE | X                | X                 |
| Num.Obs.    | 39 241           | 1864              |

+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

*Note:*

The table reports bias-corrected robust RD estimates using OLS with risk set fixed effects. Missingness in Math score occurs for all RSU students, so the estimation leverages only the UL dataset variation.

Table 3: Main Effects of Waivers on Outcomes

|             | OLS                  | Risk Set FE          | 2SLS                 | Per 1K EUR           |
|-------------|----------------------|----------------------|----------------------|----------------------|
| GPA         | 0.336***<br>(0.099)  | 0.378***<br>(0.091)  | 0.488***<br>(0.118)  | 0.160***<br>(0.046)  |
| Persistence | 0.109***<br>(0.022)  | 0.102***<br>(0.022)  | 0.133***<br>(0.029)  | 0.056***<br>(0.012)  |
| On Track    | 0.094***<br>(0.021)  | 0.089***<br>(0.022)  | 0.115***<br>(0.028)  | 0.049***<br>(0.012)  |
| Graduation  | 0.139**<br>(0.049)   | 0.124**<br>(0.041)   | 0.154**<br>(0.050)   | 0.071***<br>(0.020)  |
| Dropout     | -0.128***<br>(0.035) | -0.115***<br>(0.032) | -0.150***<br>(0.041) | -0.069***<br>(0.017) |
| On Time     | 0.171**<br>(0.054)   | 0.116*<br>(0.050)    | 0.145*<br>(0.062)    | 0.072**<br>(0.025)   |
| Risk Set FE |                      | X                    | X                    | X                    |
| Num.Obs.    | 1864                 | 1864                 | 1864                 | 1598                 |

+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

*Note:*

Note: The table reports intent-to-treat (OLS and Risk Set FE) and instrumental variable (2SLS) estimates of the effect of waivers eligibility on student outcomes. OLS estimates are from a local linear regression of the outcome on the eligibility indicator. Risk Set FE includes risk set fixed effects. 2SLS estimates instrument actual waivers receipt with eligibility. Per 1K EUR estimates instrument the monetary value of the waivers (in thousands of Euros) with eligibility. Persistence is defined as an indicator for whether the student is enrolled in the university in the following semester. On Track is defined as an indicator for whether the student progresses to the next academic semester in the following term (e.g. from semester 1 to semester 2). All models except OLS include risk set fixed effects, and all models cluster standard errors at the student level in parentheses. The minimum, maximum, and average first-stage F-statistics across the 2SLS models are 452.7, 576.3, and 525.2 respectively.

Table 4: Main Effects of Stipends on Outcomes

|             | OLS                 | Risk Set FE         | 2SLS                | Per 1K EUR          |
|-------------|---------------------|---------------------|---------------------|---------------------|
| GPA         | 0.231***<br>(0.019) | 0.214***<br>(0.019) | 0.257***<br>(0.023) | 0.507***<br>(0.045) |
| Persistence | 0.112***<br>(0.008) | 0.045***<br>(0.006) | 0.053***<br>(0.007) | 0.103***<br>(0.013) |
| On Track    | 0.164***<br>(0.009) | 0.061***<br>(0.008) | 0.072***<br>(0.009) | 0.139***<br>(0.018) |
| Graduation  | 0.154***<br>(0.010) | 0.066***<br>(0.008) | 0.076***<br>(0.009) | 0.150***<br>(0.018) |
| Dropout     | -0.028**<br>(0.010) | 0.007<br>(0.009)    | 0.012<br>(0.016)    | 0.022<br>(0.029)    |
| On Time     | 0.162***<br>(0.010) | 0.067***<br>(0.009) | 0.078***<br>(0.010) | 0.152***<br>(0.020) |
| Risk Set FE |                     | X                   | X                   | X                   |
| Num.Obs.    | 39 979              | 39 978              | 39 978              | 39 978              |

+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

*Note:*

Note: The table reports intent-to-treat (OLS and Risk Set FE) and instrumental variable (2SLS) estimates of the effect of stipends eligibility on student outcomes. OLS estimates are from a local linear regression of the outcome on the eligibility indicator. Risk Set FE includes risk set fixed effects. 2SLS estimates instrument actual stipends receipt with eligibility. Per 1K EUR estimates instrument the monetary value of the stipends (in thousands of Euros) with eligibility. Persistence is defined as an indicator for whether the student is enrolled in the university in the following semester. On Track is defined as an indicator for whether the student progresses to the next academic semester in the following term (e.g. from semester 1 to semester 2). All models except OLS include risk set fixed effects, and all models cluster standard errors at the student level in parentheses. The minimum, maximum, and average first-stage F-statistics across the 2SLS models are 2263.6, 33210.9, and 25572.9 respectively.

Table 5: Effects Over Time

|                 | GPA                 | Persistence         | On Track            | Aid                 | Aid in 000s of EUR  |
|-----------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| <i>Stipends</i> |                     |                     |                     |                     |                     |
| Next Semester   | 0.175***<br>(0.028) | 0.002<br>(0.004)    | 0.003<br>(0.004)    | 0.557***<br>(0.010) | 0.302***<br>(0.006) |
| 2 semesters out | 0.176***<br>(0.033) | 0.001<br>(0.006)    | 0.009<br>(0.006)    | 0.119***<br>(0.011) | 0.073***<br>(0.006) |
| 3 semesters out | 0.096*<br>(0.038)   | -0.017**<br>(0.006) | 0.001<br>(0.007)    | 0.064***<br>(0.012) | 0.037***<br>(0.007) |
| 4 semesters out | 0.070<br>(0.048)    | -0.012*<br>(0.006)  | 0.001<br>(0.008)    | 0.053***<br>(0.013) | 0.029***<br>(0.008) |
| 5 semesters out | 0.086<br>(0.054)    | 0.004<br>(0.005)    | 0.006<br>(0.008)    | 0.051**<br>(0.016)  | 0.031**<br>(0.010)  |
| 6 semesters out | 0.147*<br>(0.068)   | -0.003<br>(0.005)   | 0.004<br>(0.008)    | 0.032+<br>(0.018)   | 0.014<br>(0.011)    |
| Risk Set FE     | X                   | X                   | X                   | X                   | X                   |
| Num.Obs.        | 14932               | 16360               | 16212               | 16360               | 16360               |
| <i>Waivers</i>  |                     |                     |                     |                     |                     |
| Next Semester   | 0.378***<br>(0.091) | 0.102***<br>(0.022) | 0.089***<br>(0.022) | 0.769***<br>(0.023) | 2.109***<br>(0.067) |
| 2 semesters out | 0.344**<br>(0.109)  | 0.106***<br>(0.025) | 0.147***<br>(0.031) | 0.434***<br>(0.033) | 1.335***<br>(0.099) |
| 3 semesters out | 0.048<br>(0.147)    | 0.056*<br>(0.027)   | 0.151***<br>(0.032) | 0.113**<br>(0.038)  | 0.340**<br>(0.117)  |
| 4 semesters out | 0.296+<br>(0.155)   | 0.085***<br>(0.026) | 0.170***<br>(0.033) | 0.125*<br>(0.052)   | 0.324*<br>(0.140)   |
| 5 semesters out | 0.032<br>(0.229)    | 0.062**<br>(0.022)  | 0.124***<br>(0.030) | 0.146**<br>(0.055)  | 0.388**<br>(0.148)  |
| 6 semesters out | -0.059<br>(0.221)   | 0.014<br>(0.022)    | 0.114***<br>(0.030) | 0.044<br>(0.077)    | 0.133<br>(0.206)    |
| Risk Set FE     | X                   | X                   | X                   | X                   | X                   |
| Num.Obs.        | 1660                | 1864                | 1863                | 1864                | 1598                |

+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

Note: The table reports reduced-form estimates of aid eligibility on outcomes in subsequent semesters (t+1 to t+6). Each cell represents a separate regression including risk set fixed effects. Persistence is defined as an indicator for whether the student is enrolled in the university in the given semester. On Track is defined as an indicator for whether the student has progressed to the expected academic semester for that term. Robust standard errors clustered at the student level in parentheses.

Table 6: The Dynamic Aid Multiplier and Crowd-In Share: Cluster-Bootstrap Inference

| Outcome / Estimand                                   | Cash Stipends  |                 |               | Tuition Waivers |               |               |
|------------------------------------------------------|----------------|-----------------|---------------|-----------------|---------------|---------------|
|                                                      | Sem 1          | Sem 2           | Sem 3         | Sem 1           | Sem 2         | Sem 3         |
| <i>Panel A: Degree Completion (Graduation)</i>       |                |                 |               |                 |               |               |
| Total Reduced-Form Effect ( $\delta^T$ )             | 0.077          | 0.059           | 0.047         | 0.134           | 0.127         | 0.118         |
| Direct Marginal Effect ( $\delta^M$ )                | 0.062          | 0.041           | 0.040         | 0.071           | 0.140         | 0.157         |
| Dynamic Crowd-In Effect ( $\delta^{Indirect}$ )      | 0.015          | 0.018           | 0.008         | 0.064           | -0.014        | -0.040        |
| <b>Crowd-In Multiplier (<math>M</math>)</b>          | <b>1.25</b>    | <b>1.44</b>     | <b>1.20</b>   | <b>1.90</b>     | <b>0.90</b>   | <b>0.75</b>   |
|                                                      | (0.197)        | (2.256)         | (0.970)       | (3.021)         | (3.780)       | (3.700)       |
| <b>Crowd-In Share (%)</b>                            | <b>19.7%</b>   | <b>30.7%*</b>   | <b>16.6%</b>  | <b>47.3%</b>    | <b>-11.0%</b> | <b>-33.5%</b> |
|                                                      | (11.5%)        | (25.8%)         | (32.3%)       | (227.9%)        | (286.5%)      | (245.4%)      |
| <i>Panel B: Cumulative GPA (<math>\sigma</math>)</i> |                |                 |               |                 |               |               |
| Total Reduced-Form Effect ( $\delta^T$ )             | 0.199          | 0.214           | 0.193         | 0.388           | 0.214         | 0.159         |
| Direct Marginal Effect ( $\delta^M$ )                | 0.137          | 0.121           | 0.145         | 0.359           | 0.132         | 0.034         |
| Dynamic Crowd-In Effect ( $\delta^{Indirect}$ )      | 0.062          | 0.093           | 0.048         | 0.029           | 0.082         | 0.125         |
| <b>Crowd-In Multiplier (<math>M</math>)</b>          | <b>1.45**</b>  | <b>1.77**</b>   | <b>1.33*</b>  | <b>1.08</b>     | <b>1.63</b>   | <b>4.63</b>   |
|                                                      | (0.330)        | (0.967)         | (0.221)       | (2.226)         | (4.016)       | (4.272)       |
| <b>Crowd-In Share (%)</b>                            | <b>31.1%**</b> | <b>43.5%***</b> | <b>25.0%*</b> | <b>7.5%</b>     | <b>38.5%</b>  | <b>78.4%</b>  |
|                                                      | (13.9%)        | (13.1%)         | (11.4%)       | (95.1%)         | (253.1%)      | (244.8%)      |
| <i>Panel C: On-Time Graduation</i>                   |                |                 |               |                 |               |               |
| Total Reduced-Form Effect ( $\delta^T$ )             | 0.106          | 0.091           | 0.072         | 0.151           | 0.146         | 0.147         |
| Direct Marginal Effect ( $\delta^M$ )                | 0.082          | 0.059           | 0.056         | 0.091           | 0.145         | 0.115         |
| Dynamic Crowd-In Effect ( $\delta^{Indirect}$ )      | 0.024          | 0.031           | 0.016         | 0.061           | 0.001         | 0.033         |
| <b>Crowd-In Multiplier (<math>M</math>)</b>          | <b>1.30*</b>   | <b>1.52**</b>   | <b>1.28</b>   | <b>1.67</b>     | <b>1.00</b>   | <b>1.29</b>   |
|                                                      | (0.187)        | (0.686)         | (0.717)       | (3.709)         | (2.426)       | (3.444)       |
| <b>Crowd-In Share (%)</b>                            | <b>23.1%</b>   | <b>34.4%***</b> | <b>22.0%</b>  | <b>40.1%</b>    | <b>0.5%</b>   | <b>22.2%</b>  |
|                                                      | (11.9%)        | (37.5%)         | (26.1%)       | (253.0%)        | (289.1%)      | (264.3%)      |

Note: This table computes the dynamic aid multiplier and crowd-in share across academic semesters from college entry. The crowd-in multiplier measures the factor by which the direct marginal effects are expanded by crowded in financial aid, and the crowd-in share captures the proportion of the total effect attributable to future crowded-in aid. Standard errors in parentheses are computed from 500 student-level cluster bootstrap replications. Significance stars denote rejection of the null hypothesis of no dynamic crowd-in (multiplier equal to 1.00 and share equal to 0.0%): \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$  based on the empirical bootstrap distribution.

Table 7: Dynamic Complementarity: Effects on Next Semester GPA

|                             | Current CGPA       | Past Aid            | Math HS (UL)     |
|-----------------------------|--------------------|---------------------|------------------|
| <i>Stipends</i>             |                    |                     |                  |
| Aid Receipt                 | 0.018<br>(0.027)   | 0.214***<br>(0.034) | 0.151<br>(0.115) |
| Aid $\times$ Current CGPA   | 0.061*<br>(0.025)  |                     |                  |
| Aid $\times$ Past Aid (Lag) |                    | 0.039<br>(0.039)    |                  |
| Aid $\times$ Math HS (UL)   |                    |                     | 0.000<br>(0.001) |
| Risk Set FE                 | X                  | X                   | X                |
| Num.Obs.                    | 27386              | 18847               | 5902             |
| <i>Waivers</i>              |                    |                     |                  |
| Aid Receipt                 | 0.293**<br>(0.098) | 0.301*<br>(0.126)   |                  |
| Aid $\times$ Current CGPA   | 0.010<br>(0.083)   |                     |                  |
| Aid $\times$ Past Aid (Lag) |                    | 0.009<br>(0.206)    |                  |
| Risk Set FE                 | X                  | X                   |                  |
| Num.Obs.                    | 1502               | 1140                |                  |

+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

*Note:*

Results from two-stage least squares regressions estimating the impact of current aid receipt on next-semester GPA. The endogenous interactions test for dynamic complementarities across proxies of skill and resources for both stipends and waivers. All models include risk set fixed effects and control for the running variable. Robust standard errors clustered at the student level.

Table 8: Dynamic Complementarity: Effects on Graduation

|                             | Current CGPA         | Past Aid            | Math HS (UL)      |
|-----------------------------|----------------------|---------------------|-------------------|
| <i>Stipends</i>             |                      |                     |                   |
| Aid Receipt                 | 0.077***<br>(0.011)  | 0.070***<br>(0.013) | 0.088+<br>(0.053) |
| Aid $\times$ Current CGPA   | -0.040***<br>(0.010) |                     |                   |
| Aid $\times$ Past Aid (Lag) |                      | -0.024+<br>(0.015)  |                   |
| Aid $\times$ Math HS (UL)   |                      |                     | 0.000<br>(0.001)  |
| Risk Set FE                 | X                    | X                   | X                 |
| Num.Obs.                    | 33177                | 22725               | 8062              |
| <i>Waivers</i>              |                      |                     |                   |
| Aid Receipt                 | 0.084*<br>(0.042)    | 0.149**<br>(0.047)  |                   |
| Aid $\times$ Current CGPA   | -0.136**<br>(0.043)  |                     |                   |
| Aid $\times$ Past Aid (Lag) |                      | -0.041<br>(0.073)   |                   |
| Risk Set FE                 | X                    | X                   |                   |
| Num.Obs.                    | 1137                 | 970                 |                   |

+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

*Note:*

Results from two-stage least squares regressions estimating the impact of current aid receipt on the ultimate probability of graduation. The endogenous interactions test for dynamic complementarities across proxies of skill and resources for both stipends and waivers. All models include risk set fixed effects and control for the running variable. Robust standard errors clustered at the student level.

Table 9: Heterogeneity by Demographics in the Effects of Aid on Student Outcomes

| Outcome         | Heterogeneity Variable |                   |                   |                    |
|-----------------|------------------------|-------------------|-------------------|--------------------|
|                 | Male                   | Age               | Minority          | Capital            |
| <i>Stipends</i> |                        |                   |                   |                    |
| GPA             | -0.095+<br>(0.050)     | -0.001<br>(0.004) | -0.003<br>(0.057) | -0.002<br>(0.039)  |
| Persistence     | 0.018<br>(0.015)       | 0.001<br>(0.001)  | 0.019<br>(0.017)  | 0.031**<br>(0.012) |
| On Track        | 0.021<br>(0.020)       | 0.001<br>(0.001)  | 0.035<br>(0.024)  | 0.033*<br>(0.016)  |
| Dropout         | 0.031<br>(0.033)       | -0.001<br>(0.002) | -0.039<br>(0.029) | -0.012<br>(0.018)  |
| Graduation      | -0.011<br>(0.020)      | 0.002<br>(0.002)  | 0.028<br>(0.024)  | 0.013<br>(0.016)   |
| Ontime          | -0.003<br>(0.022)      | -0.001<br>(0.002) | 0.056*<br>(0.027) | 0.024<br>(0.018)   |
| Risk Set FE     | X                      | X                 | X                 | X                  |
| Num.Obs.        | 39241                  | 38891             | 38113             | 38113              |
| <i>Waivers</i>  |                        |                   |                   |                    |
| GPA             | -0.013<br>(0.291)      | 0.008<br>(0.015)  | -0.277<br>(0.283) | 0.016<br>(0.184)   |
| Persistence     | -0.076<br>(0.061)      | 0.005<br>(0.004)  | 0.064<br>(0.072)  | 0.001<br>(0.047)   |
| On Track        | -0.066<br>(0.061)      | 0.005<br>(0.004)  | 0.062<br>(0.072)  | 0.004<br>(0.045)   |
| Dropout         | 0.108<br>(0.083)       | 0.002<br>(0.005)  | -0.040<br>(0.088) | 0.050<br>(0.064)   |
| Graduation      | 0.003<br>(0.107)       | -0.004<br>(0.006) | 0.034<br>(0.097)  | -0.015<br>(0.080)  |
| Ontime          | 0.024<br>(0.128)       | -0.007<br>(0.006) | -0.004<br>(0.147) | 0.024<br>(0.093)   |
| Risk Set FE     | X                      | X                 | X                 | X                  |
| Num.Obs.        | 1864                   | 1827              | 1827              | 1827               |

+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

<sup>a</sup> The table reports heterogeneity analysis by student demographics. The coefficients reflect the interaction between aid eligibility and the respective demographic indicator. All models include risk set fixed effects. Robust standard errors clustered at the student level.

Table 10: Heterogeneity by Cutoff Characteristics in the Effects of Aid on Student Outcomes

| Outcome         | Heterogeneity Variable |                      |                      |                      |                      |
|-----------------|------------------------|----------------------|----------------------|----------------------|----------------------|
|                 | Cutoff Percentile      | First Year           | First Half           | Last Year            | Last Semester        |
| <i>Stipends</i> |                        |                      |                      |                      |                      |
| GPA             | -0.189+<br>(0.103)     | -0.154***<br>(0.039) | -0.143***<br>(0.037) | 0.050<br>(0.037)     | 0.055<br>(0.044)     |
| Persistence     | -0.048<br>(0.031)      | 0.035**<br>(0.013)   | 0.025*<br>(0.011)    | 0.003<br>(0.012)     | 0.006<br>(0.014)     |
| On Track        | -0.041<br>(0.035)      | 0.017<br>(0.015)     | 0.014<br>(0.015)     | 0.013<br>(0.015)     | 0.040*<br>(0.019)    |
| Dropout         | 0.048<br>(0.040)       | -0.002<br>(0.023)    | -0.013<br>(0.016)    | -0.016<br>(0.017)    | 0.021<br>(0.020)     |
| Graduation      | -0.069+<br>(0.040)     | 0.014<br>(0.017)     | 0.013<br>(0.015)     | 0.021<br>(0.015)     | 0.031<br>(0.019)     |
| Ontime          | -0.067<br>(0.042)      | -0.001<br>(0.017)    | 0.016<br>(0.016)     | 0.016<br>(0.016)     | 0.025<br>(0.021)     |
| Risk Set FE     | X                      | X                    | X                    | X                    | X                    |
| Num.Obs.        | 39978                  | 39978                | 39945                | 39945                | 39945                |
| <i>Waivers</i>  |                        |                      |                      |                      |                      |
| GPA             | -0.498<br>(0.378)      | 0.258<br>(0.182)     | 0.091<br>(0.190)     | 0.146<br>(0.182)     | 0.272<br>(0.304)     |
| Persistence     | -0.308**<br>(0.095)    | 0.002<br>(0.047)     | 0.075+<br>(0.040)    | -0.152***<br>(0.040) | 0.004<br>(0.059)     |
| On Track        | -0.312***<br>(0.091)   | 0.016<br>(0.046)     | 0.066+<br>(0.040)    | -0.129***<br>(0.039) | -0.004<br>(0.063)    |
| Dropout         | 0.109<br>(0.131)       | -0.025<br>(0.070)    | -0.127*<br>(0.057)   | 0.064<br>(0.063)     | 0.139*<br>(0.059)    |
| Graduation      | -0.230<br>(0.180)      | 0.017<br>(0.093)     | 0.199*<br>(0.078)    | 0.065<br>(0.082)     | -0.304***<br>(0.086) |
| Ontime          | -0.275<br>(0.217)      | -0.074<br>(0.115)    | 0.048<br>(0.097)     | 0.043<br>(0.112)     | -0.142<br>(0.132)    |
| Risk Set FE     | X                      | X                    | X                    | X                    | X                    |
| Num.Obs.        | 1864                   | 1864                 | 1864                 | 1864                 | 1864                 |

+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

Note: The table reports combined heterogeneity analysis by characteristics of the cutoff itself. Each column reports the coefficient on the interaction term between eligibility and the characteristic. Cutoff Percentile is the percentile rank of the cutoff within the risk set. First Half indicates if the student is in the first half of their program. All models include risk set fixed effects. Robust standard errors clustered at the student level.

Table 11: Heterogeneity by Competitiveness in the Effects of Aid on Student Outcomes

| Outcome         | Heterogeneity Variable |                   |                     |                      |
|-----------------|------------------------|-------------------|---------------------|----------------------|
|                 | Fall (RSU)             | <Median GPA       | <Median Progression | Guaranteed Aid (RSU) |
| <i>Stipends</i> |                        |                   |                     |                      |
| GPA             | -0.030<br>(0.053)      | 0.005<br>(0.038)  | 0.032<br>(0.039)    | -0.292*<br>(0.129)   |
| Persistence     | -0.012<br>(0.008)      | 0.009<br>(0.012)  | 0.057***<br>(0.010) | 0.004<br>(0.014)     |
| On Track        | -0.012<br>(0.008)      | -0.012<br>(0.015) | 0.086***<br>(0.014) | 0.003<br>(0.014)     |
| Dropout         | -0.003<br>(0.014)      | 0.028<br>(0.018)  |                     | -0.023<br>(0.037)    |
| Graduation      | 0.019<br>(0.017)       | -0.001<br>(0.015) | 0.069***<br>(0.014) | -0.002<br>(0.057)    |
| On time         | 0.027<br>(0.024)       | -0.014<br>(0.017) | 0.048**<br>(0.017)  | 0.004<br>(0.073)     |
| Risk Set FE     | X                      | X                 | X                   | X                    |
| Num.Obs.        | 16360                  | 39978             | 39978               | 16360                |
| <i>Waivers</i>  |                        |                   |                     |                      |
| GPA             | 0.145<br>(0.196)       | 0.350+<br>(0.184) | -0.091<br>(0.180)   |                      |
| Persistence     | -0.090+<br>(0.051)     | 0.089*<br>(0.044) | 0.113*<br>(0.045)   |                      |
| On Track        | -0.063<br>(0.048)      | 0.076+<br>(0.043) | 0.100*<br>(0.043)   |                      |
| Dropout         | 0.053<br>(0.072)       | -0.049<br>(0.057) | 0.053<br>(0.061)    |                      |
| Graduation      | -0.069<br>(0.082)      | 0.049<br>(0.070)  | -0.006<br>(0.076)   |                      |
| On time         | -0.006<br>(0.107)      | -0.037<br>(0.088) | -0.003<br>(0.095)   |                      |
| Risk Set FE     | X                      | X                 | X                   | X                    |
| Num.Obs.        | 1864                   | 1864              | 1864                | 0                    |

+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

<sup>a</sup> The table reports combined heterogeneity analysis by various measures of competitiveness. Each column reports the coefficient on the interaction term between aid eligibility and the column variable (*Above*  $\times$  *Het*). All models include risk set fixed effects and allow separate linear trends by subgroup on both sides of the cutoff (*Above*  $\times$  *Het*  $\times$  *Runvar*). Fall (RSU) indicates Fall semester, estimated only for RSU students who receive a waiver in the Fall for two semesters. <Median GPA indicates below-median baseline GPA. <Median Progression indicates higher difficulty risk sets based on a predetermined, leave-one-cohort-out historical on-track rate for the same study program and semester. Guaranteed Aid (RSU) indicates students who started with a waiver, also estimated only for RSU. Robust standard errors clustered at the student level in parentheses.

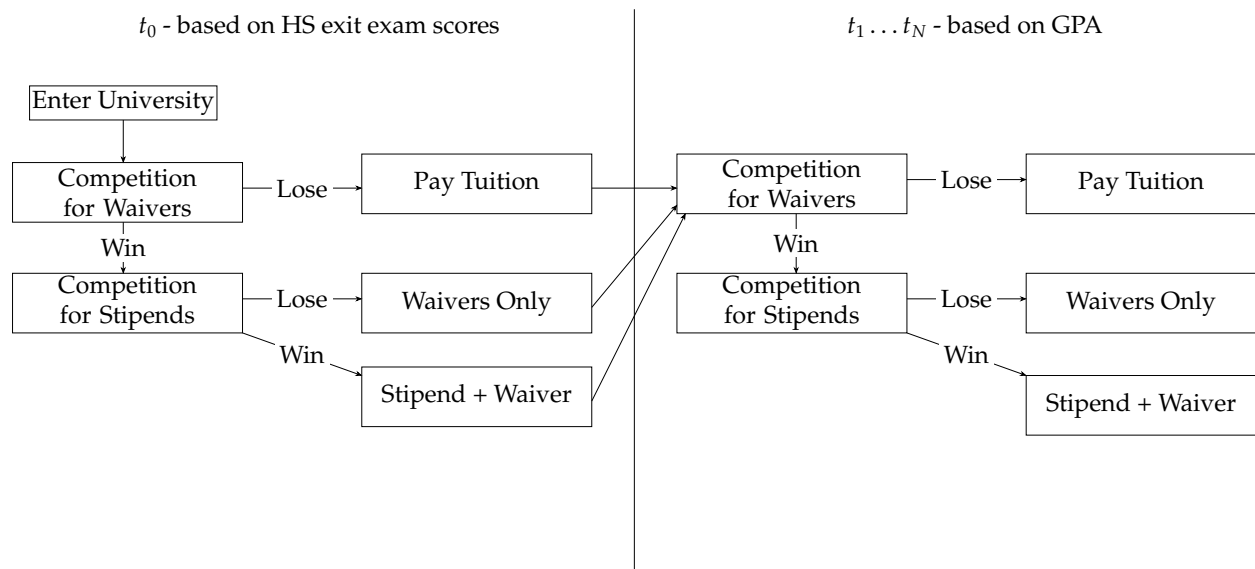
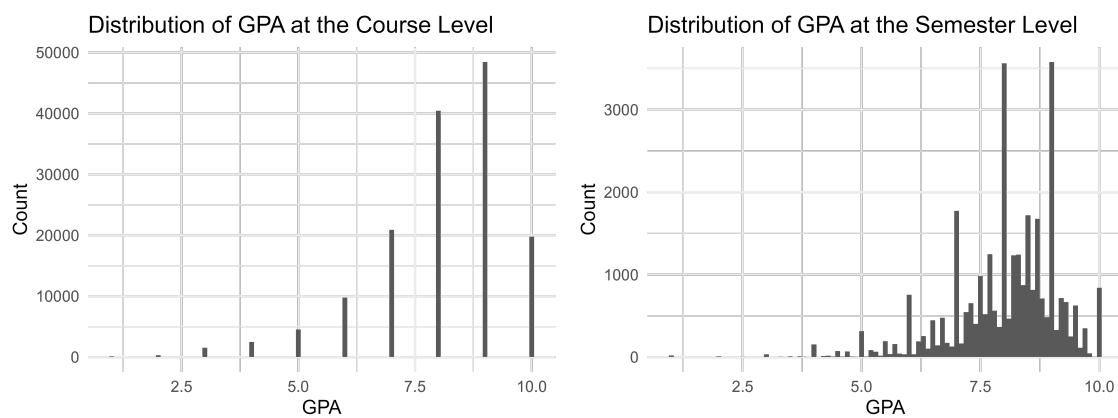


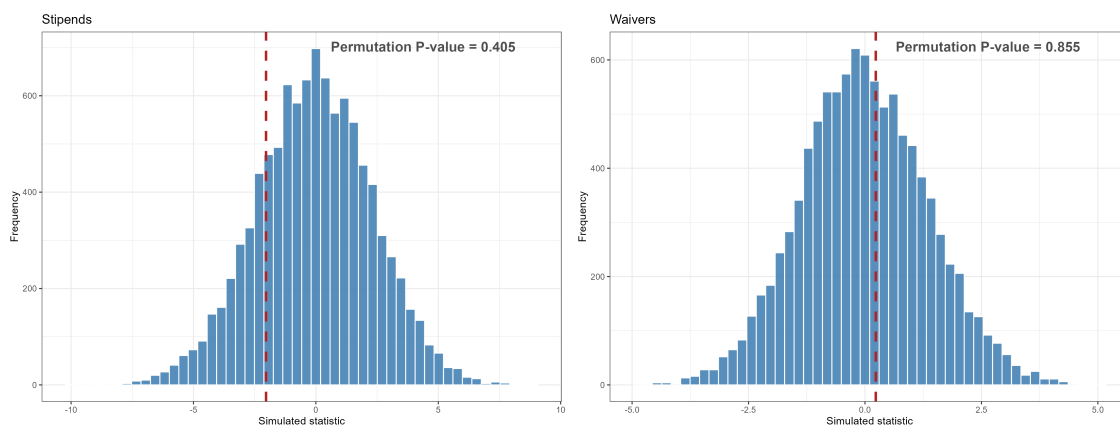
Figure 1: The Student Financial Aid Assignment Mechanism in a Dual-Track Tuition Model

Figure 2: Raw GPA at the Course and Semester Levels



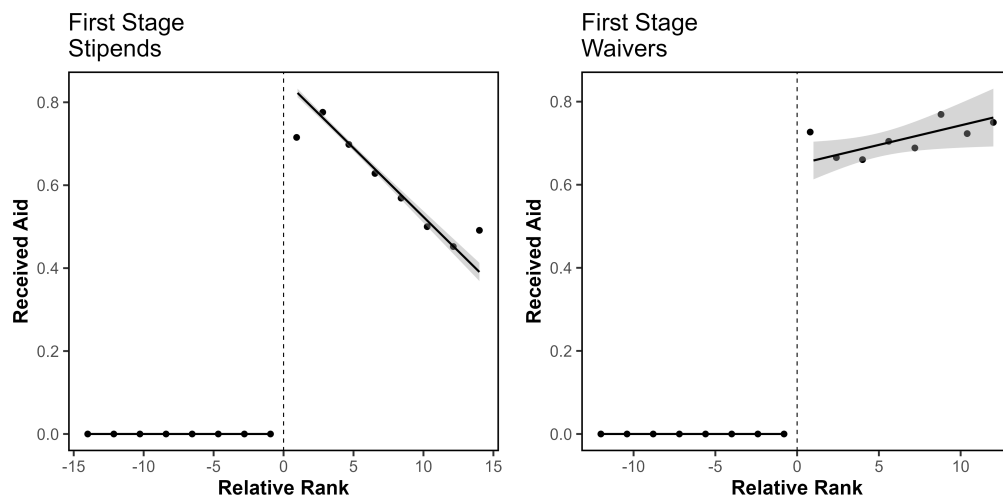
*Note:* The figure on the left shows the raw GPA that students acquire at the course level, and the figure on the right shows the average semester GPA that is used to assign financial aid at RSU.

Figure 3: Distribution of Density Test Values for Stipends and Waivers



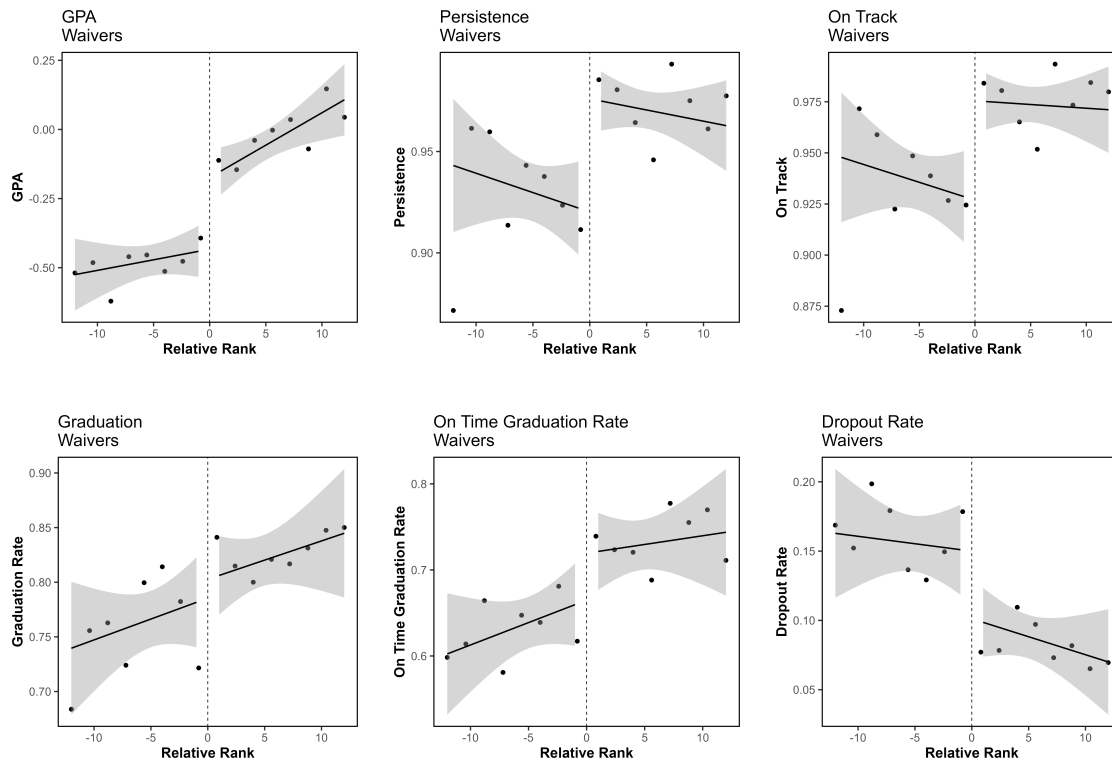
*Note:* These figures show the distribution of density test values for the dataset wherein the cutoff values are randomly reassigned within risk sets. Line indicates density test value for observed data.

Figure 4: Change in Aid Receipt Probability for Stipends and Waivers



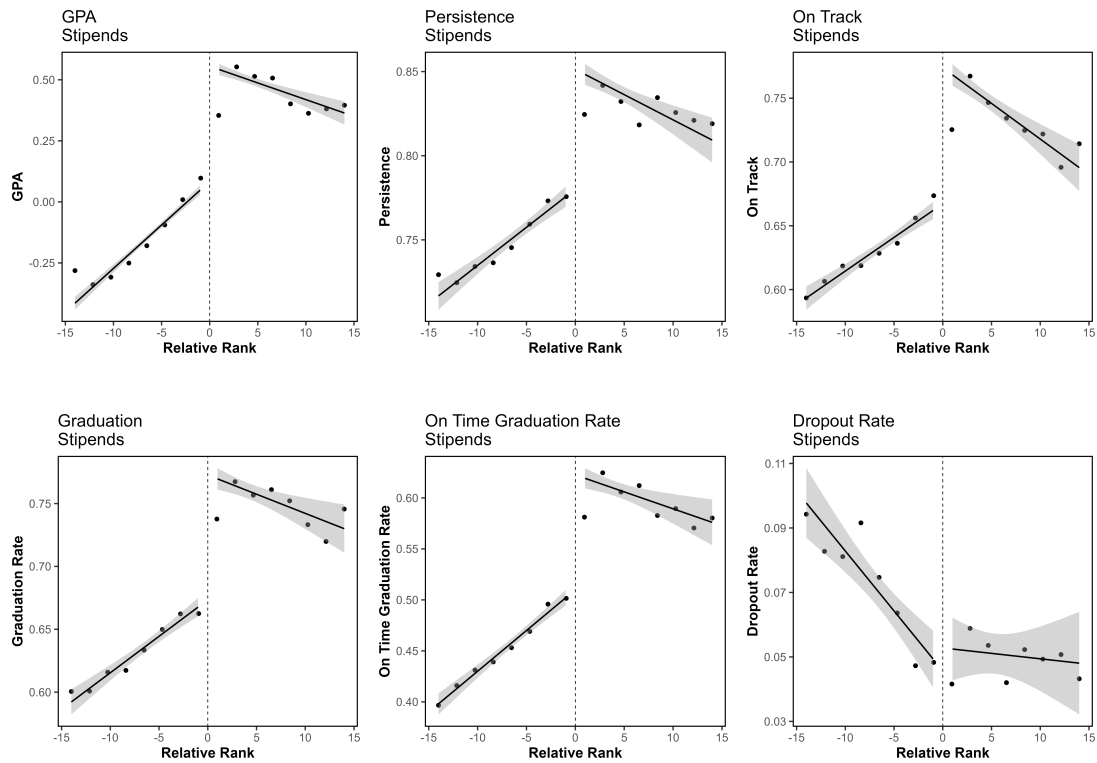
*Note:* These figures show the change in student probability of aid receipt based on rank distance from the individual with the smallest GPA to have received a given type of financial aid. The plotted points represent binned averages of the raw data, and the lines are linear fits to the raw data separately for each side of the cutoff.

Figure 5: Student Outcomes Below and Above the Tuition Waiver Reception Threshold



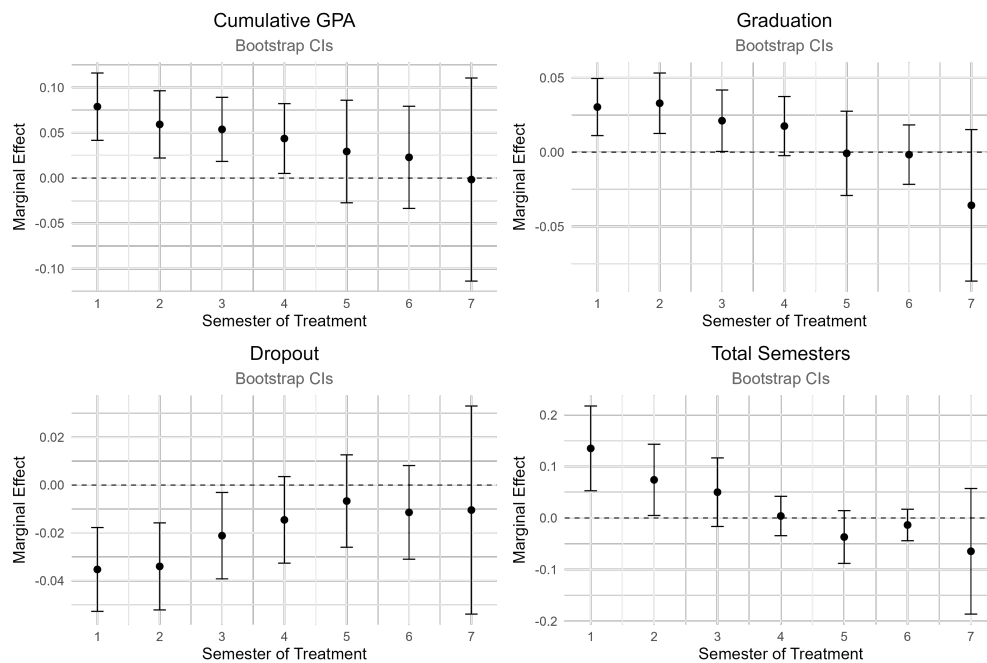
*Note:* These figures show the change in student outcomes based on distance from the rank cutoff for reception of tuition waivers. The plotted points are binned averages of the dependent variable, residualized by risk-set fixed effects. The lines are linear fits to the residualized data separately for each side of the cutoff.

Figure 6: Student Outcomes Below and Above the Stipend Reception Threshold



*Note:* These figures show the change in student outcomes based on distance from the rank cutoff for the reception of stipends across both institutions. The plotted points are binned averages of the dependent variable, residualized by risk-set fixed effects. The lines are linear fits to the residualized data separately for each side of the cutoff. Dropout is omitted as it is only tracked in the RSU subset.

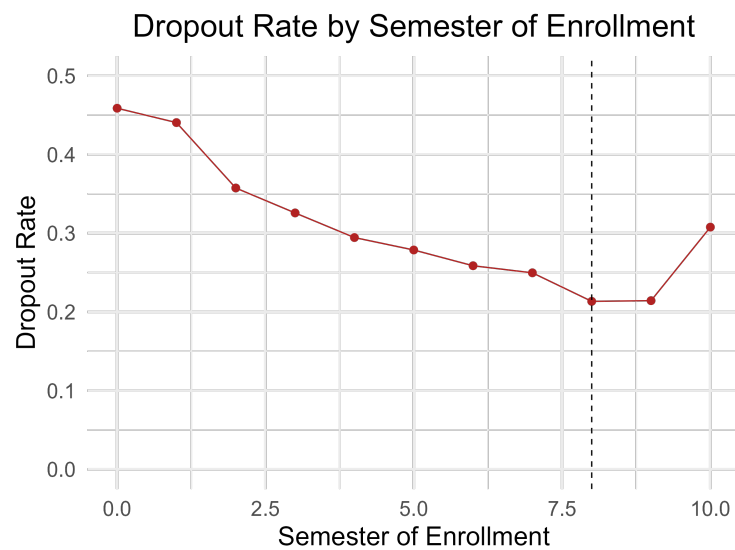
Figure 7: Marginal Effect of Waiver Eligibility at a Given Semester



*Note:* The figures report the estimates from an indirect regression discontinuity decomposition on the marginal impact of being eligible for waivers and total received aid in any given semester on a range of student outcomes - cumulative GPA, graduation, ever dropping out, and the total number of semesters that a student is enrolled for. The estimation is done on students who start during the time period and whose expected graduation falls before the data horizon, and whose program length is 8 semesters.

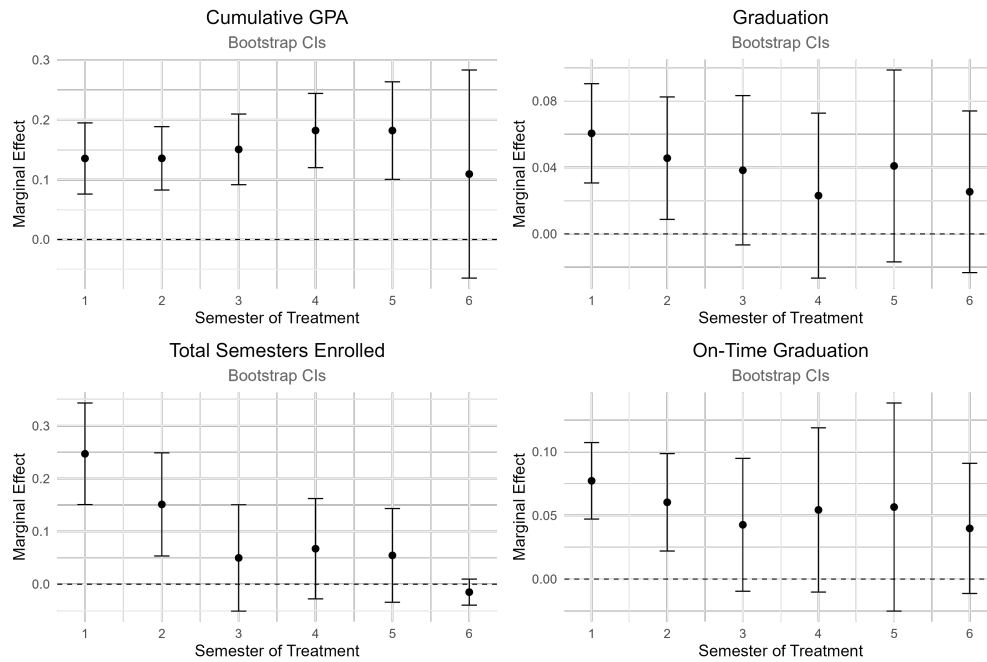
Standard errors acquired by conducting a cluster bootstrap.

Figure 8: Probability of Dropping out by Semester for Undergraduates With a Program Duration of 4 years



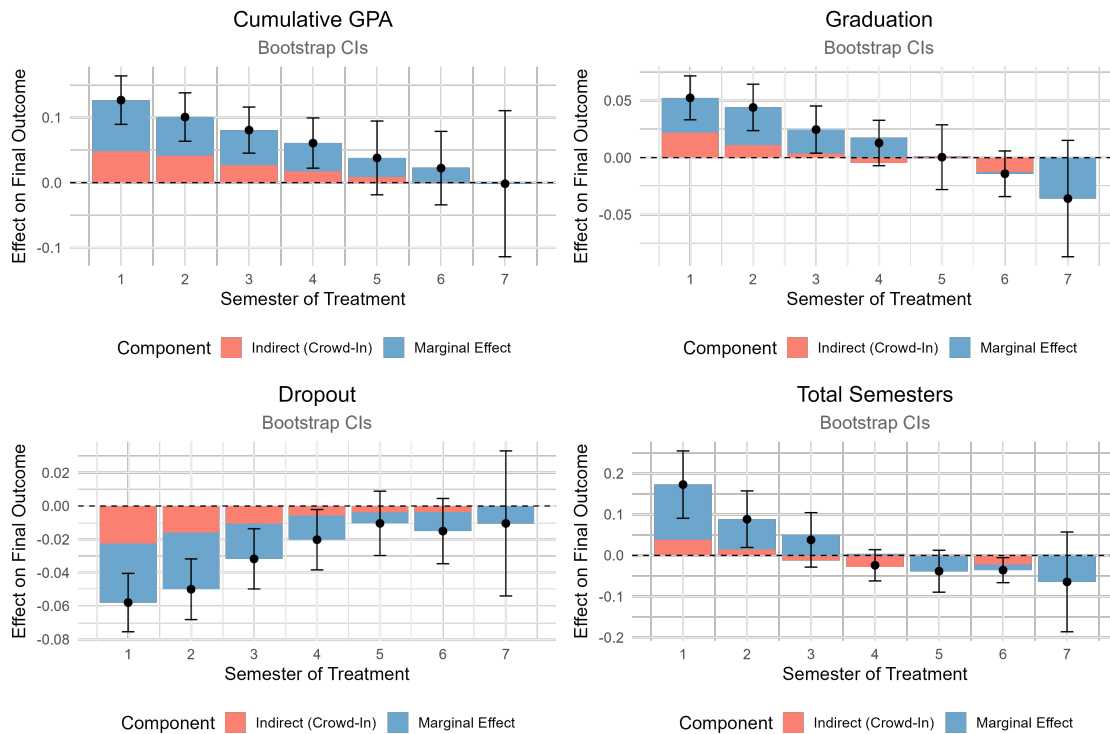
*Note:* This figure reports the probability that an individual observed in a given semester would drop out at any future time for a subset of individuals who are studying in 4-year undergraduate programs. Dashed line indicates semester 8 - the final semester in one's studies for this subgroup.

Figure 9: Marginal Effect of Stipend Eligibility at a Given Semester



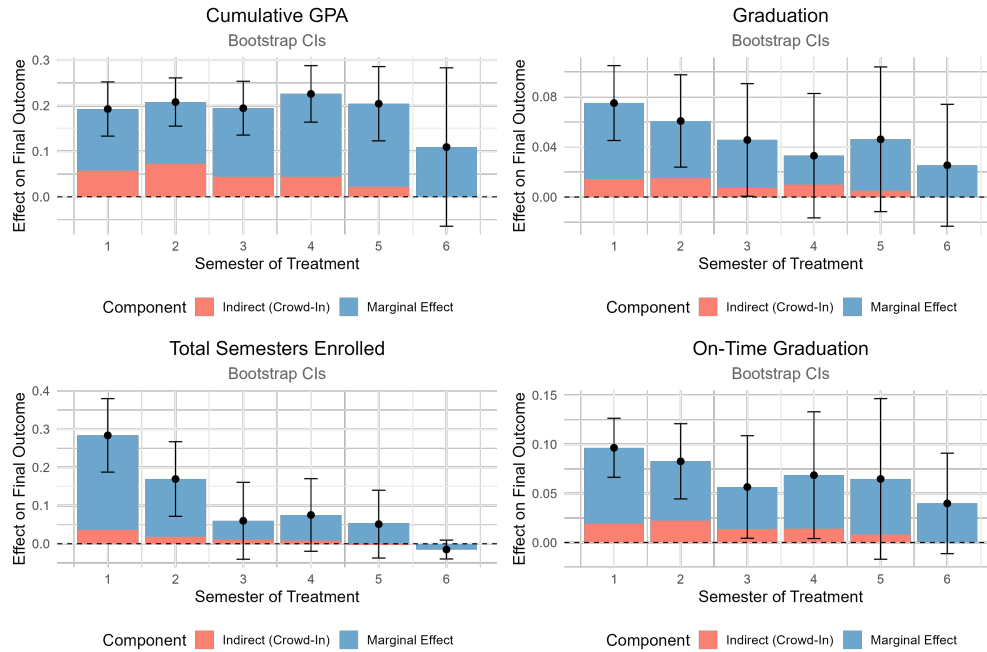
*Note:* The figures report the estimates from an indirect regression discontinuity decomposition on the marginal impact of being eligible for stipends in any given semester on a range of student outcomes. The estimation is done on students who start during the time period and whose expected graduation falls before the data horizon, and whose program length is 6 semesters.

Figure 10: Decomposition of Impact of Waiver Eligibility

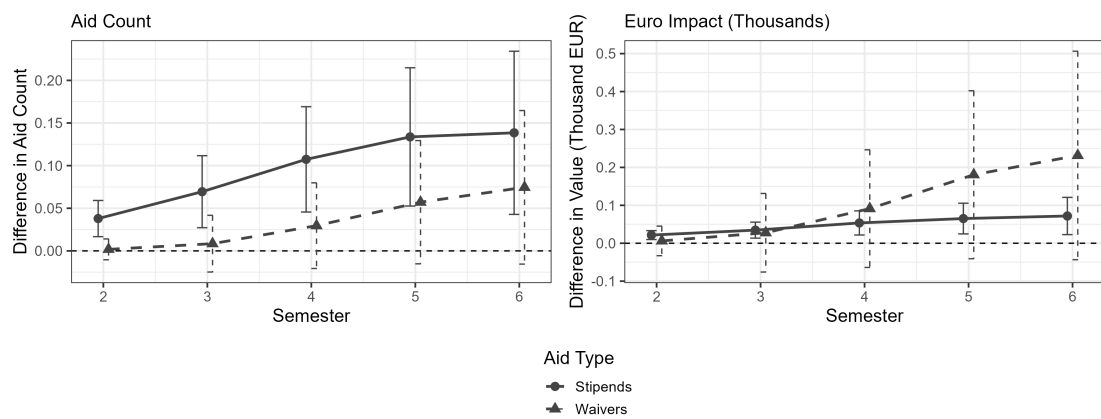


*Note:* The figures report the estimates from an indirect regression discontinuity decomposition on the marginal, indirect, and total impact of being eligible for waivers in any given semester on a range of student outcomes. The estimation is done on students who start during the time period and whose expected graduation falls before the data horizon, and whose program length is 8 semesters. Standard errors acquired using cluster bootstrap.

Figure 11: Decomposition of Impact of Stipend Eligibility

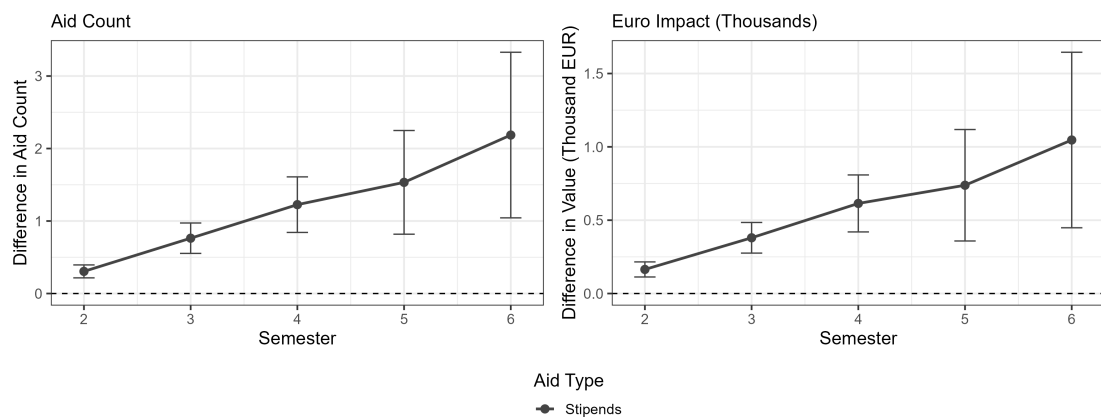


*Note:* The figures report the estimates from an indirect regression discontinuity decomposition on the marginal, indirect, and total impact of being eligible for stipends in any given semester on a range of student outcomes. The estimation is done on students who start during the time period and whose expected graduation falls before the data horizon, and whose program length is 6 semesters. Standard errors estimated through cluster bootstrap.



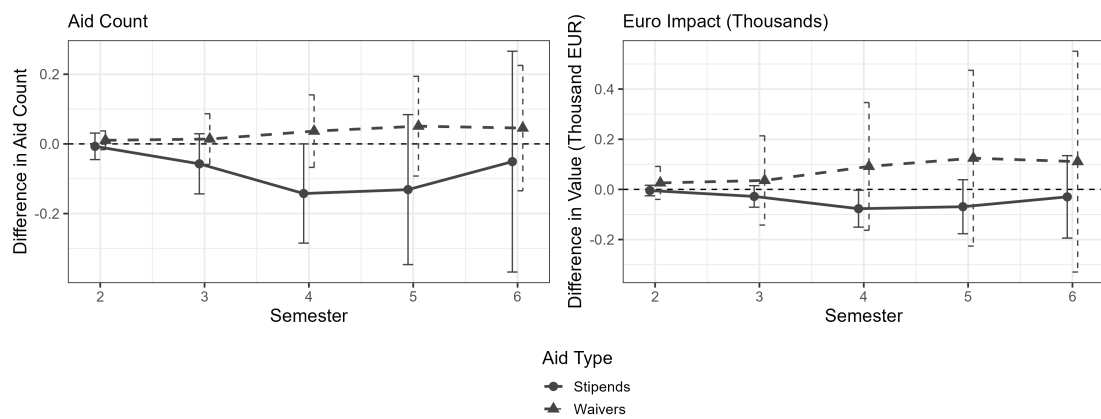
Omitted Category: Non-Capital

Figure 12: Cumulative Aid Divergence by Having Finished High School in the Capital



Omitted Category: Math Q1, Semester 1 (Estimates shown for Math Q5)

Figure 13: Cumulative Aid Divergence by High School Math Exit Exam Score



Omitted Category: Male

Figure 14: Cumulative Aid Divergence by Gender

# Appendices

## A Density Tests for Sparse Data

The main assumption behind an RD design is that student potential outcomes are continuous across the cutoff, i.e. that the only student characteristics that suddenly change across the cutoff are their probability of treatment receipt. To support this assumption, there are multiple tests, with the two most prominent being a balance test of student characteristics (reported in Table 2), and a density test of whether there is differential heaping around the cutoff (Cattaneo et al., 2020; McCrary, 2008).

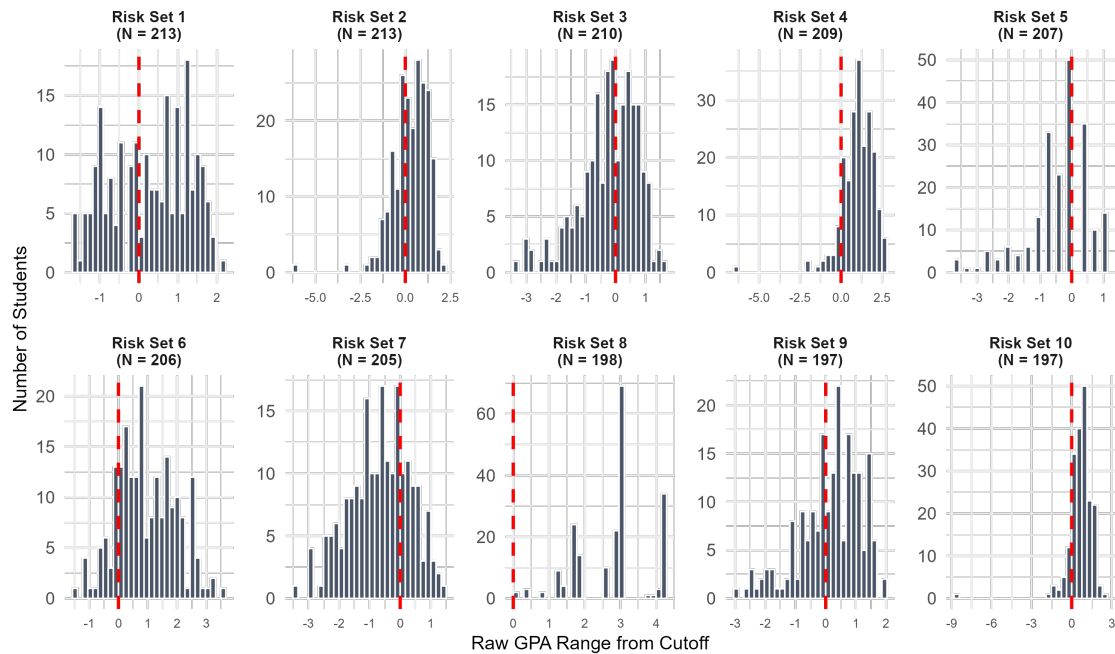
The core concern that these tests address is whether students are able to select into treatment or not. If students are able to select into treatment, then the RD estimator will be biased. The balance test looks at observable characteristics and sees whether those change discontinuously, and the density test looks at the heaping around the cutoff to see if people are differentially selecting into treatment, which would show up as a large number of individuals being just below or above the cutoff.

The way my data is constructed is that students take courses in a semester that can only take on full values ranging from 1 to 10, with the semester grade being the average of these values. Because of the way the data is generated, regular density tests would yield incorrect inference, because the discontinuous distributions could just as well be attributed to the data-generating process as to student selection into treatment.

The tournament-style provision of financial aid means that for students to be able to select into treatment, they would need to push another student out, which would inherently balance the densities across the cutoff. However, the null hypothesis of no discontinuity is rejected by the density tests due to the data-generating process.

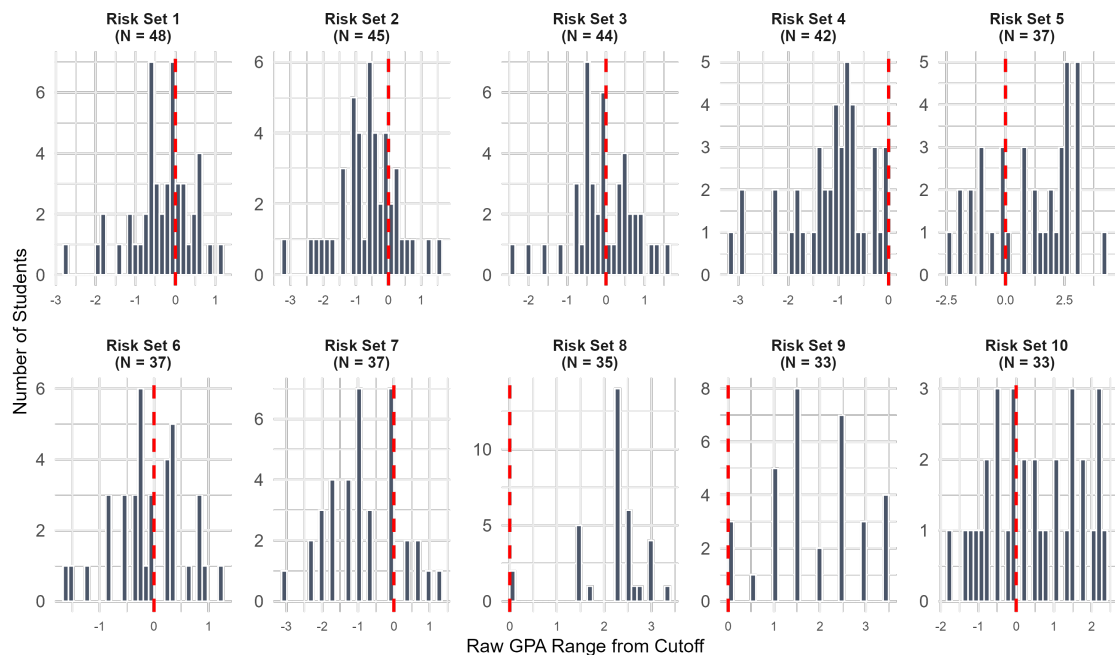
I plot the 10 largest risk sets for stipends (Figure 15) and waivers (Figure 16). Thus, when looking at each risk set individually and at the raw scores, I do not find evidence of

Figure 15: Density of Raw GPA Scores for 10 Largest Stipend Risk Sets



*Note:* These plots represent the density of the raw GPA of individuals in the 10 largest risk sets. The GPA has been recentered so that 0 is the cutoff at which the individual with the lowest GPA received aid.

Figure 16: Density of Raw GPA Scores for 10 Largest Waiver Risk Sets



*Note:* These plots represent the density of the raw GPA of individuals in the 10 largest risk sets. The GPA has been recentered so that 0 is the cutoff at which the individual with the lowest GPA received aid.

individuals being able to precisely manipulate their GPA and self-select into aid reception.

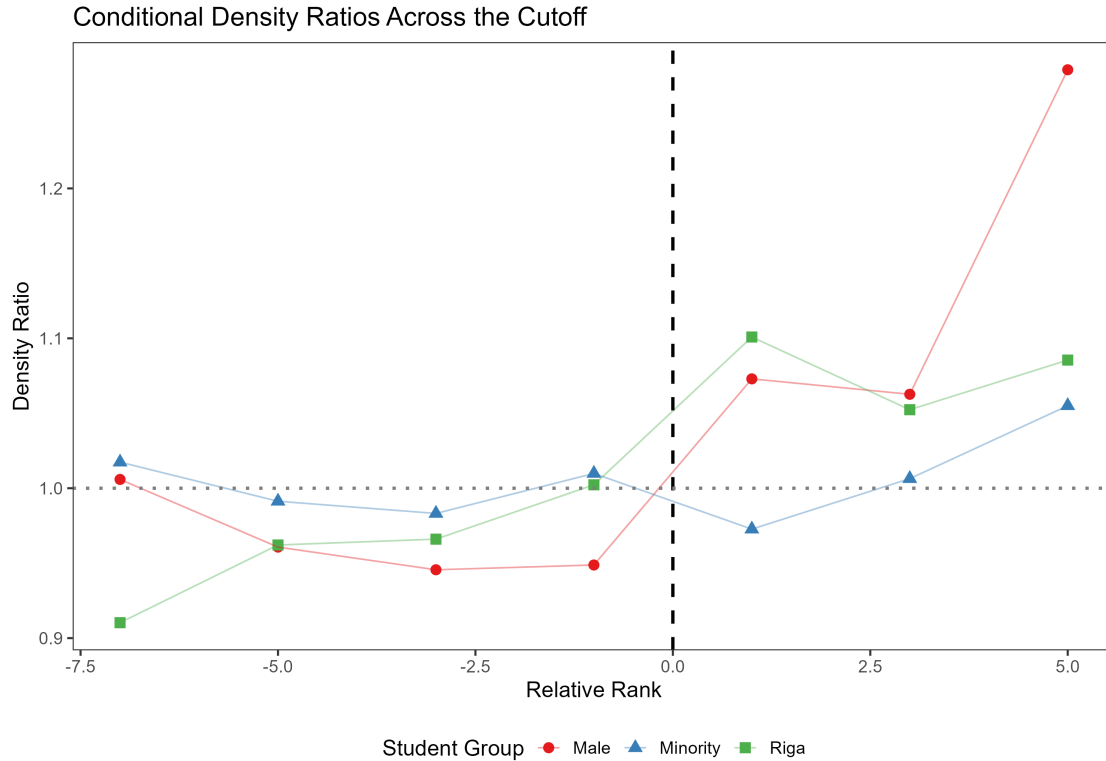


Figure 17: Conditional Density Ratio Test

To empirically estimate the degree to which there could be selective discontinuities across the cutoff, I use two different tests - a test for the discontinuity in the ratios of the conditional densities to the unconditional density (Zimmerman, 2014), and a permutation test. The permutation test is reported in the main section of the paper.

The conditional density ratio test looks at the conditional distribution of student densities across the cutoffs to see if they are continuous across the cutoff. The test looks at not only the continuous distribution of standardized grades across the cutoff, but of student conditional densities based on characteristics that impact their academic outcomes. I report the results of this test in Figure 17 for three different groups - male students, students from minority-language-of-instruction schools, and students from Riga. I find that each of the density ratios is continuous around the cutoff value.

Table 12: Estimation of Stipend Eligibility Effects with Different Bandwidths for RSU

|                    | Bandwidth           |                     |                     |                     |                     |
|--------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                    | (1)                 | (2)                 | (3)                 | (4)                 | (5)                 |
| GPA                | 0.132***<br>(0.039) | 0.179***<br>(0.028) | 0.179***<br>(0.028) | 0.229***<br>(0.026) | 0.326***<br>(0.023) |
| Persistence        | 0.002<br>(0.006)    | 0.002<br>(0.004)    | 0.002<br>(0.004)    | 0.000<br>(0.004)    | 0.000<br>(0.003)    |
| On Track           | -0.001<br>(0.006)   | 0.003<br>(0.004)    | 0.003<br>(0.004)    | 0.000<br>(0.003)    | 0.001<br>(0.003)    |
| Graduation         | 0.004<br>(0.014)    | -0.004<br>(0.010)   | -0.004<br>(0.010)   | -0.005<br>(0.009)   | 0.010<br>(0.008)    |
| Dropout            | -0.002<br>(0.011)   | 0.011<br>(0.008)    | 0.011<br>(0.008)    | 0.008<br>(0.007)    | 0.000<br>(0.006)    |
| Graduation on Time | 0.046*<br>(0.018)   | 0.025+<br>(0.013)   | 0.025+<br>(0.013)   | 0.007<br>(0.013)    | 0.021+<br>(0.012)   |
| Risk Set FE        | X                   | X                   | X                   | X                   | X                   |
| Num.Obs.           | 9967                | 15 352              | 15 352              | 18 412              | 21 868              |

+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

Note: The table reports reduced-form estimates of stipend eligibility estimated using varying bandwidths of the running variable (relative rank). Column headers indicate the bandwidth size in rank positions. Persistence is defined as an indicator for whether the student is enrolled in the university in the following semester. All models include risk set fixed effects. Robust standard errors clustered at the student level in parentheses.

## B Robustness Tests

### B.1 Varying the Bandwidth

I also conduct robustness tests by varying the bandwidth (tables 13 and 12). I find that my results are robust to different bandwidth specifications.

### B.2 Varying the Cutoff Point

I also conduct robustness tests by varying the point at which the cutoff occurs and similarly find that centering the cutoff at 0 provides the strongest predicted probability of treatment.

Table 13: Estimation of Waiver Eligibility Effects with Different Bandwidths

|                    | Bandwidth           |                    |                      |                      |                      |
|--------------------|---------------------|--------------------|----------------------|----------------------|----------------------|
|                    | (1)                 | (2)                | (3)                  | (4)                  | (5)                  |
| GPA                | 0.326+<br>(0.193)   | 0.357+<br>(0.208)  | 0.420***<br>(0.091)  | 0.378***<br>(0.101)  | 0.418***<br>(0.075)  |
| Persistence        | 0.107**<br>(0.038)  | 0.095+<br>(0.049)  | 0.107***<br>(0.022)  | 0.079**<br>(0.025)   | 0.091***<br>(0.017)  |
| On Track           | 0.070+<br>(0.039)   | 0.072<br>(0.046)   | 0.094***<br>(0.022)  | 0.065**<br>(0.024)   | 0.081***<br>(0.017)  |
| Graduation         | 0.221*<br>(0.085)   | 0.267**<br>(0.088) | 0.132**<br>(0.042)   | 0.148***<br>(0.043)  | 0.087*<br>(0.036)    |
| Dropout            | -0.188**<br>(0.060) | -0.167*<br>(0.072) | -0.130***<br>(0.033) | -0.131***<br>(0.035) | -0.090***<br>(0.027) |
| Graduation on Time | 0.148<br>(0.107)    | 0.235*<br>(0.110)  | 0.122*<br>(0.050)    | 0.113*<br>(0.057)    | 0.102*<br>(0.041)    |
| Risk Set FE        | X                   | X                  | X                    | X                    | X                    |
| Num.Obs.           | 458                 | 771                | 1592                 | 1418                 | 1952                 |

+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

Note: The table reports reduced-form estimates of waiver eligibility estimated using varying bandwidths of the running variable (relative rank). Column headers indicate the bandwidth size in rank positions. Persistence is defined as an indicator for whether the student is enrolled in the university in the following semester. All models include risk set fixed effects. Robust standard errors clustered at the student level in parentheses.

Table 14: Estimation of First Stage with Various Cutoffs for RSU

|                 | Cutoff Value         |                      |                     |                     |                     |
|-----------------|----------------------|----------------------|---------------------|---------------------|---------------------|
|                 | -2                   | -1                   | 0                   | 1                   | 2                   |
| <i>Stipends</i> |                      |                      |                     |                     |                     |
| Stipend         | -0.076***<br>(0.016) | 0.028+<br>(0.016)    | 0.557***<br>(0.010) | 0.449***<br>(0.010) | 0.304***<br>(0.010) |
| Num.Obs.        | 15401                | 15978                | 16360               | 16705               | 17075               |
| <i>Waivers</i>  |                      |                      |                     |                     |                     |
| Waiver          | -0.325***<br>(0.041) | -0.140***<br>(0.039) | 0.769***<br>(0.023) | 0.508***<br>(0.030) | 0.243***<br>(0.028) |
| Num.Obs.        | 1776                 | 1826                 | 1864                | 1854                | 1778                |
| Risk Set FE     | X                    | X                    | X                   | X                   | X                   |

+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

Note: The table reports first-stage estimates using placebo cutoffs (artificially shifting the cutoff rank). The dependent variable is aid receipt. Column headers indicate the shift in the cutoff rank. All models include risk set fixed effects. Robust standard errors clustered at the student level in parentheses.

Table 15: Estimation of Stipend Eligibility Effects with Different Bandwidths for UL

|                    | Bandwidth           |                     |                     |                     |                     |
|--------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                    | (1)                 | (2)                 | (3)                 | (4)                 | (5)                 |
| GPA                | 0.187***<br>(0.027) | 0.204***<br>(0.025) | 0.285***<br>(0.021) | 0.364***<br>(0.019) | 0.410***<br>(0.018) |
| Persistence        | 0.064***<br>(0.009) | 0.066***<br>(0.008) | 0.061***<br>(0.007) | 0.065***<br>(0.006) | 0.073***<br>(0.006) |
| On Track           | 0.071***<br>(0.012) | 0.076***<br>(0.011) | 0.093***<br>(0.009) | 0.103***<br>(0.009) | 0.119***<br>(0.008) |
| Graduation         | 0.086***<br>(0.011) | 0.088***<br>(0.010) | 0.087***<br>(0.009) | 0.093***<br>(0.008) | 0.103***<br>(0.007) |
| Graduation on Time | 0.064***<br>(0.012) | 0.070***<br>(0.011) | 0.092***<br>(0.009) | 0.103***<br>(0.009) | 0.119***<br>(0.008) |
| Risk Set FE        | X                   | X                   | X                   | X                   | X                   |
| Num.Obs.           | 21 907              | 24 220              | 30 922              | 35 760              | 38 629              |

+ p < 0.1, \* p < 0.05, \*\* p < 0.01, \*\*\* p < 0.001

Note: The table reports the estimates from regressions from a reduced-form regression for various bandwidths. Robust standard errors clustered at the student level in parentheses.

Table 16: Estimation of First Stage with Various Cutoffs for UL

|             | Cutoff Value         |                     |                     |                     |                     |
|-------------|----------------------|---------------------|---------------------|---------------------|---------------------|
|             | -2                   | -1                  | 0                   | 1                   | 2                   |
| Stipend     | -0.250***<br>(0.007) | 0.157***<br>(0.008) | 0.949***<br>(0.004) | 0.541***<br>(0.007) | 0.182***<br>(0.006) |
| Risk Set FE | X                    | X                   | X                   | X                   | X                   |
| Num.Obs.    | 21 201               | 22 828              | 24 231              | 25 326              | 26 125              |

+ p < 0.1, \* p < 0.05, \*\* p < 0.01, \*\*\* p < 0.001

Note: The table reports the estimates from regressions of student of receiving aid on whether they are above the cutoff or not for various cutoff values with risk set fixed effects. Robust standard errors clustered at the student level in parentheses.

I also conduct these tests for UL and find that the results are also robust.

### B.3 Using Two-Way Clustering

In the main estimation, I cluster the standard errors at the student level to account for serial correlation across semesters. Because students compete within risk sets and have common time-based and class-based shocks, it could be that their standard errors are also serially correlated across risk sets. To account for this, I conduct estimates using two-way clustering by both student and risk sets and report them in table 17. I find that clustering by both student and risk set does not meaningfully change the results.

### B.4 Local Randomization Inference

An additional concern that might occur is that the local linear estimates are driven by functional form assumptions or by ties of students at rank 0. To test these concerns, I estimate the effects using non-parametric local randomization (Cattaneo et al., 2017). Table 18 reports the results.

Column 1 compares students at rank +1 directly against students at rank -1 within the same risk set, dropping boundary rank 0 entirely (a discrete donut design). Column 2 evaluates the minimal single-rank jump from rank 0 to rank -1, and Column 3 evaluates a narrow window of two ranks on each side ( $\{-2, -1\}$  versus  $\{0, +1\}$ ). Across all comparisons, degree completion effects remain large and statistically indistinguishable from baseline estimates (+8.6pp to +11.9pp, for stipends; +12.2pp to +17.9pp for waivers), proving that results do not depend on functional form extrapolation.

### B.5 Using the Armstrong and Kolesár (2020) Approach

Finally, I also use the estimation approach by Armstrong and Kolesár (2020) to conduct tests using their suggested RD estimation approach. I let the bandwidth vary for each outcome variable and aid type and estimate the reduced-form estimates in Table 19 and 2SLS estimates in Table 20. I generally find broadly similar outcomes as with my main

Table 17: Robustness to Two-Way Clustering: Student and Risk-Set Level Dependence

| Outcome Margin                  | Reduced Form (ITT)            |                               | 2SLS (Local Wald Ratio)       |                               |
|---------------------------------|-------------------------------|-------------------------------|-------------------------------|-------------------------------|
|                                 | 1-Way (Student)               | 2-Way (Risk Set)              | 1-Way (Student)               | 2-Way (Risk Set)              |
| <i>Panel A: Cash Stipends</i>   |                               |                               |                               |                               |
| Next-Semester GPA ( $\sigma$ )  | 0.183***<br>(0.020)           | 0.183***<br>(0.021)           | 0.218***<br>(0.024)           | 0.218***<br>(0.025)           |
| Persistence to Next Semester    | 0.048***<br>(0.006)           | 0.048***<br>(0.006)           | 0.057***<br>(0.007)           | 0.057***<br>(0.007)           |
| On-Track Academic Progression   | 0.056***<br>(0.008)           | 0.056***<br>(0.008)           | 0.066***<br>(0.010)           | 0.066***<br>(0.010)           |
| Degree Completion (Graduation)  | 0.068***<br>(0.008)           | 0.068***<br>(0.008)           | 0.078***<br>(0.010)           | 0.078***<br>(0.010)           |
| Dropout from University         | 0.007<br>(0.010)              | 0.007<br>(0.010)              | 0.013<br>(0.017)              | 0.013<br>(0.018)              |
| On-Time Degree Completion       | 0.063***<br>(0.009)           | 0.063***<br>(0.009)           | 0.073***<br>(0.011)           | 0.073***<br>(0.011)           |
| <i>Panel B: Tuition Waivers</i> |                               |                               |                               |                               |
| Next-Semester GPA ( $\sigma$ )  | 0.346***<br>(0.101)           | 0.346***<br>(0.102)           | 0.431***<br>(0.126)           | 0.431***<br>(0.129)           |
| Persistence to Next Semester    | 0.077**<br>(0.025)            | 0.077**<br>(0.028)            | 0.096**<br>(0.031)            | 0.096**<br>(0.035)            |
| On-Track Academic Progression   | 0.062**<br>(0.024)            | 0.062*<br>(0.026)             | 0.078**<br>(0.030)            | 0.078*<br>(0.033)             |
| Degree Completion (Graduation)  | 0.139**<br>(0.042)            | 0.139**<br>(0.042)            | 0.166***<br>(0.050)           | 0.166**<br>(0.050)            |
| Dropout from University         | -0.119***<br>(0.034)          | -0.119***<br>(0.035)          | -0.149***<br>(0.042)          | -0.149***<br>(0.043)          |
| On-Time Degree Completion       | 0.101 <sup>†</sup><br>(0.056) | 0.101 <sup>†</sup><br>(0.059) | 0.121 <sup>†</sup><br>(0.067) | 0.121 <sup>†</sup><br>(0.071) |
| Risk Set Fixed Effects          | Yes                           | Yes                           | Yes                           | Yes                           |
| Running Variable Controls       | Linear                        | Linear                        | Linear                        | Linear                        |
| Cluster Variable 1              | Student                       | Student                       | Student                       | Student                       |
| Cluster Variable 2              | None                          | Risk Set                      | None                          | Risk Set                      |

Note: This table evaluates the sensitivity of the main baseline regression discontinuity estimates (Table 4 and Table 3) to within-tournament dependence. In Column 1 and Column 3, standard errors are clustered at the individual student level (`apl_nr`), following the baseline specification. In Column 2 and Column 4, standard errors are clustered two-way by both student and risk set (`apl_nr + risk_set`), accounting for common shocks, cohort-level grading differences, and zero-sum assignment coupling within competition groups. Clustering two-way by student and risk set  $\times$  semester yields identical results to three decimal places.

<sup>†</sup>  $p < 0.10$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ .

specification.

Table 18: Local Randomization Inference at Discrete Adjacent Ranks

| Outcome Margin                                     | Adjacent Pair<br>Ranks $\{+1\}$ vs. $\{-1\}$ | Boundary Step<br>Ranks $\{0\}$ vs. $\{-1\}$ | Narrow Band<br>$\{-2, -1\}$ vs. $\{0, 1\}$ |
|----------------------------------------------------|----------------------------------------------|---------------------------------------------|--------------------------------------------|
| <i>Panel A: Cash Stipends (Combined Sample)</i>    |                                              |                                             |                                            |
| Next-Semester GPA ( $\sigma$ )                     | 0.447***<br>(0.032)                          | 0.240***<br>(0.026)                         | 0.372***<br>(0.017)                        |
| Persistence to Next Semester                       | 0.088***<br>(0.010)                          | 0.063***<br>(0.008)                         | 0.072***<br>(0.005)                        |
| On-Track Academic Progression                      | 0.096***<br>(0.013)                          | 0.073***<br>(0.010)                         | 0.088***<br>(0.007)                        |
| Degree Completion (Graduation)                     | 0.107***<br>(0.012)                          | 0.086***<br>(0.010)                         | 0.096***<br>(0.007)                        |
| Dropout from University                            | -0.003<br>(0.017)                            | -0.024*<br>(0.011)                          | -0.025**<br>(0.008)                        |
| On-Time Degree Completion                          | 0.107***<br>(0.013)                          | 0.088***<br>(0.011)                         | 0.101***<br>(0.008)                        |
| <i>Panel B: Cash Stipends (UL GPA-Only Design)</i> |                                              |                                             |                                            |
| Next-Semester GPA ( $\sigma$ )                     | 0.445***<br>(0.035)                          | 0.233***<br>(0.031)                         | 0.390***<br>(0.021)                        |
| Persistence to Next Semester                       | 0.099***<br>(0.011)                          | 0.078***<br>(0.010)                         | 0.091***<br>(0.007)                        |
| On-Track Academic Progression                      | 0.108***<br>(0.015)                          | 0.092***<br>(0.013)                         | 0.113***<br>(0.009)                        |
| Degree Completion (Graduation)                     | 0.119***<br>(0.013)                          | 0.100***<br>(0.012)                         | 0.114***<br>(0.008)                        |
| On-Time Degree Completion                          | 0.120***<br>(0.014)                          | 0.091***<br>(0.013)                         | 0.110***<br>(0.009)                        |
| <i>Panel C: Tuition Waivers (RSU)</i>              |                                              |                                             |                                            |
| Next-Semester GPA ( $\sigma$ )                     | 0.614***<br>(0.174)                          | 0.411***<br>(0.113)                         | 0.522***<br>(0.076)                        |
| Persistence to Next Semester                       | 0.085*<br>(0.039)                            | 0.095***<br>(0.026)                         | 0.088***<br>(0.020)                        |
| On-Track Academic Progression                      | 0.085*<br>(0.039)                            | 0.075**<br>(0.025)                          | 0.077***<br>(0.019)                        |
| Degree Completion (Graduation)                     | 0.165**<br>(0.059)                           | 0.179***<br>(0.044)                         | 0.122***<br>(0.030)                        |
| Dropout from University                            | -0.126*<br>(0.059)                           | -0.150***<br>(0.036)                        | -0.117***<br>(0.025)                       |
| On-Time Degree Completion                          | 0.194*<br>(0.078)                            | 0.173**<br>(0.059)                          | 0.146**<br>(0.045)                         |
| Risk Set Fixed Effects                             | Yes                                          | Yes                                         | Yes                                        |
| Two-Way Clustered SEs                              | Student + Risk Set                           | Student + Risk Set                          | Student + Risk Set                         |
| Boundary Rank 0 Excluded                           | Yes (Donut)                                  | No                                          | No                                         |

*Note:* This table presents discrete local randomization inference following Cattaneo et al. (2017) within narrow discrete rank windows adjacent to the tournament boundary. Column 1 compares students at rank +1 directly against students at rank -1 within the same risk set, dropping the boundary rank 0 entirely (a discrete donut design). Column 2 evaluates the minimal single-step jump from rank 0 to rank -1. Column 3 evaluates a narrow band of two ranks on each side of the cutoff (ranks  $\{-2, -1\}$  versus  $\{0, 1\}$ ). All models condition on risk set fixed effects and report two-way clustered standard errors by student and risk set. <sup>†</sup> $p < 0.10$ , \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ .

Table 19: Effect of Aid on Student Outcomes with Armstrong and Kolesar (2020) CIs

|                 | GPA                 | Persistence         | On Track            | Graduation          | Dropout            | On Time             |
|-----------------|---------------------|---------------------|---------------------|---------------------|--------------------|---------------------|
| <i>Stipends</i> |                     |                     |                     |                     |                    |                     |
| RD Estimate     | 0.151***<br>(0.023) | 0.048***<br>(0.006) | 0.056***<br>(0.008) | 0.069***<br>(0.008) | 0.012<br>(0.007)   | 0.063***<br>(0.009) |
| <i>Waivers</i>  |                     |                     |                     |                     |                    |                     |
| RD Estimate     | 0.297*<br>(0.128)   | 0.083**<br>(0.027)  | 0.063*<br>(0.026)   | 0.132**<br>(0.044)  | -0.174*<br>(0.059) | 0.104<br>(0.054)    |

+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

*Note:*

The table reports honest regression discontinuity estimates and 95% Fixed-Length Confidence Intervals (FLCI) following Armstrong and Kolesár (2020) and Kolesár and Rothe (2018), accounting for worst-case bias and discrete running variable support. Potential outcomes are assumed to belong to a second-order Hölder class  $\mathcal{H}_2(M)$  bounding the second derivative ( $|f''(r)| \leq M$ ). Curvature bounds  $M$  are estimated via Armstrong and Kolesár's data-driven rule-of-thumb (ROT) global polynomial procedure: for Stipends, GPA ( $M = 0.0071$ ,  $h = 4$ ), Persistence ( $M = 0.0007$ ,  $h = 6$ ), On Track ( $M = 0.0007$ ,  $h = 6$ ), Graduation ( $M = 0.0010$ ,  $h = 5$ ), Dropout ( $M = 0.0004$ ,  $h = 9$ ), On Time ( $M = 0.0011$ ,  $h = 5$ ); for Waivers, GPA ( $M = 0.0446$ ,  $h = 3$ ), Persistence ( $M = 0.0071$ ,  $h = 4$ ), On Track ( $M = 0.0064$ ,  $h = 4$ ), Graduation ( $M = 0.0081$ ,  $h = 4$ ), Dropout ( $M = 0.0363$ ,  $h = 2$ ), On Time ( $M = 0.0122$ ,  $h = 4$ ). Bandwidths  $h$  are MSE-optimal under the minimax worst-case bias bound using a uniform kernel. Confidence intervals use folded-normal critical values ( $cv \in [2.19, 2.60]$ ) that account for the maximum worst-case bias ratio  $b/se$ . Following Kolesár and Rothe (2018), discrete integer ranks are accommodated directly because the bound  $M$  limits the maximum linear extrapolation bias between adjacent discrete mass points (e.g., between rank  $-1$  and  $0$ ). All models include risk set fixed effects.

Table 20: 2SLS Estimates of the Effect of Aid with Armstrong and Kolesar (2020) CIs

|                        | GPA                 | Persistence         | On Track            | Graduation          | Dropout          | On Time             |
|------------------------|---------------------|---------------------|---------------------|---------------------|------------------|---------------------|
| <i>Stipends (2SLS)</i> |                     |                     |                     |                     |                  |                     |
| RD Estimate            | 0.177***<br>(0.027) | 0.057***<br>(0.007) | 0.066***<br>(0.009) | 0.080***<br>(0.009) | 0.021<br>(0.013) | 0.073***<br>(0.010) |

+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

*Note:* The table reports honest 2SLS regression discontinuity estimates and 95% Fixed-Length Confidence Intervals (FLCI) following Armstrong and Kolesár (2020), where scholarship receipt is instrumented by eligibility at the tournament cutoff. Potential outcomes belong to a second-order Hölder class  $\mathcal{H}_2(M_{\text{RF}}, M_{\text{FS}})$ . Curvature bounds  $M$  are estimated via the Armstrong and Kolesár (2020) rule-of-thumb procedure: for Stipends, GPA ( $M = 0.0086, h = 4$ ), Persistence ( $M = 0.0009, h = 6$ ), On Track ( $M = 0.0010, h = 6$ ), Graduation ( $M = 0.0012, h = 5$ ), Dropout ( $M = 0.0006, h = 9$ ), On Time ( $M = 0.0013, h = 5$ ). Bandwidths  $h$  are MSE-optimal under the minimax worst-case bias bound using a uniform kernel. Confidence intervals use folded-normal critical values ( $cv \in [2.20, 2.63]$ ) reflecting worst-case reduced-form and first-stage bias. Following Kolesár and Rothe (2018), the Hölder bound limits extrapolation bias across discrete integer ranks. All models include risk set fixed effects.

## C The One-Step Dynamic RD Method

As a complementary approach, I also adapt the one-step dynamic RD estimator (Cellini et al., 2010) to estimate the dynamic impact profile of past eligibility events on current outcomes. The difference between the one-step estimate and the recursive estimate would be that the recursive estimate described in the previous section is that the recursive estimate estimates the marginal contribution of aid in different semesters on a student's outcome.

The one-step estimator, on the other hand, estimates the impact of aid reception after time  $t$ , without being able to decompose the impact into direct and indirect effects. Another way to think about it would be to think of the one-step estimator as estimating the decay pattern of financial aid, while holding aid history constant.

The benefits of the one-step dynamic RD approach is that it uses the full history of observations for each individual, thus increasing statistical power. It also estimates the impact of financial aid over time. Using this method, I estimate the following model using panel data:

$$Y_{ijt} = \beta_0 \text{Above}_{ijt} + f_0(r_{ijt}) + \sum_{\tau=1}^{\bar{\tau}} [\beta_{\tau} \text{Above}_{ijt-\tau} + f_{\tau}(r_{ijt-\tau})] + \lambda_j + \omega_t + \epsilon_{ijt} \quad (11)$$

where  $Above_{ijt-\tau}$  indicates eligibility status  $\tau$  semesters prior,  $f_{\tau}(r_{ijt-\tau})$  is a flexible function of the rank from  $\tau$  semesters prior,  $\lambda_j$  represents risk-set fixed effects, and  $\omega_t$  denotes semester fixed effects. The coefficients  $\beta_{\tau}$  estimate the average effect for individuals above the cutoff  $\tau$  periods in the past on the current outcome  $Y_{ijt}$ , conditional on the history of eligibility and rank included in the model. Plotting  $\beta_{\tau}$  against the lag  $\tau$  directly illustrates the impact over time, or decay pattern, of initial eligibility. While estimating a different parameter than the marginal effect  $\delta_{t,t+k}^M$  from the recursive method, this approach provides valuable insights into effect duration and will offer greater statistical precision for estimating the persistence pattern itself, as it will leverage the full dataset.

Additionally, for the one-step estimator I use the same subset of individuals as I use for the dynamic decomposition - people in undergraduate programs with a program length of 8 semesters.

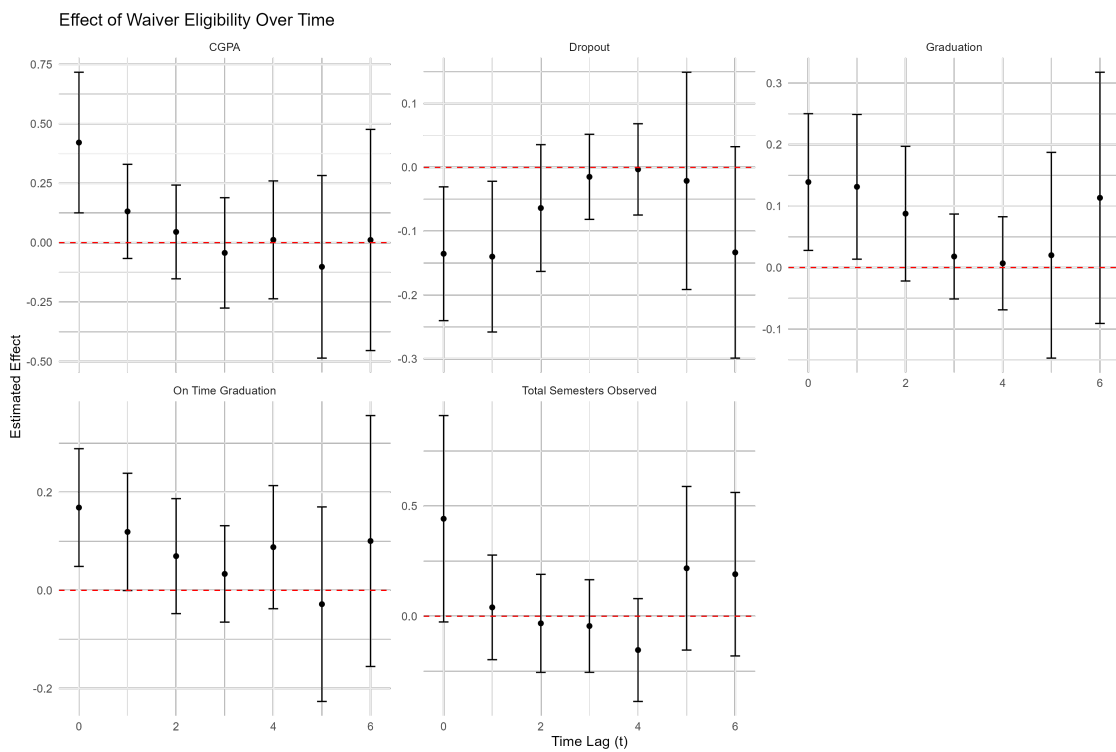
I find that even when controlling for aid history, financial aid in past semesters has strong effects on outcomes in the current semesters, with waivers having impact two semesters ahead (Figure 18) and stipends having extensive and pronounced impact on GPA throughout the observed history (Figure 19)

Additionally, I explore the heterogeneity of financial aid impact decay by conditions of the competition. As waivers received in the fall are guaranteed for two semesters, I study how financial aid receipt in the fall compared to the spring impacts the effects over time in Figures 20 and 21. I generally do not find differential impacts of having more stable waivers for students.

Finally, given that some students have guaranteed waivers throughout their studies if they received a waiver upon enrolment, I conduct a heterogeneity analysis for stipend impacts by interacting stipend aid receipt with whether the student had a guaranteed waiver or not (Figure 22).

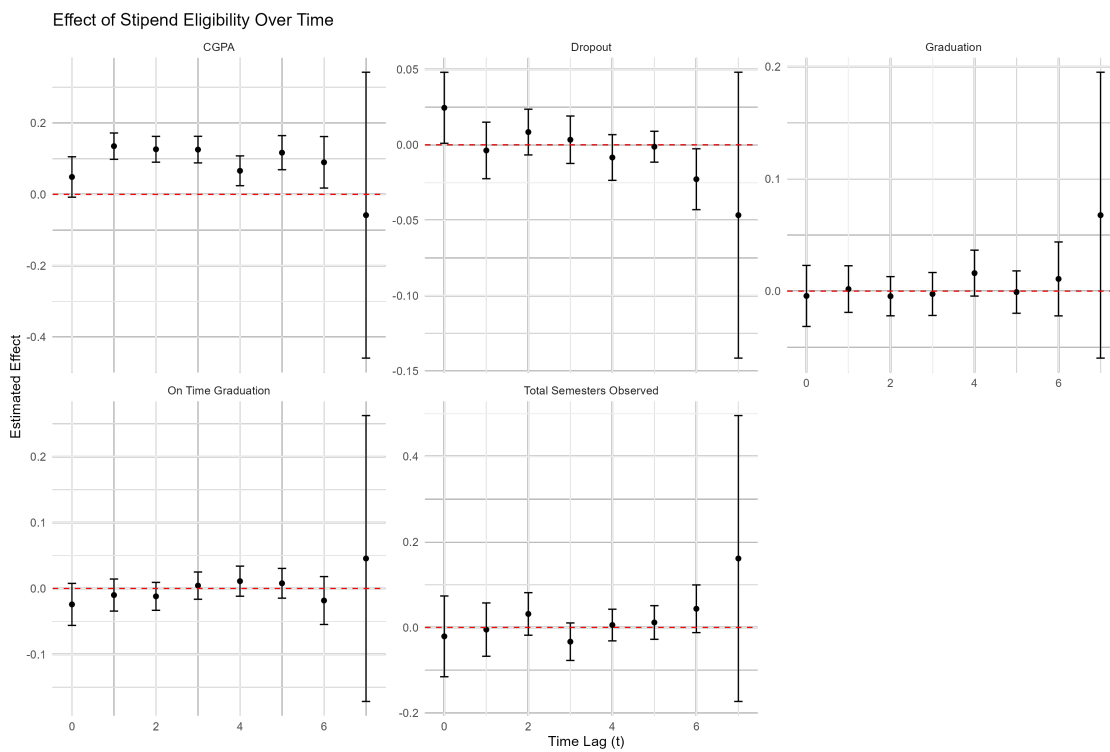
I find that individuals who received stipends while in guaranteed waiver seats see smaller impacts on their GPA over time, potentially driven by the security effect of having

Figure 18: One-shot estimator results for waivers



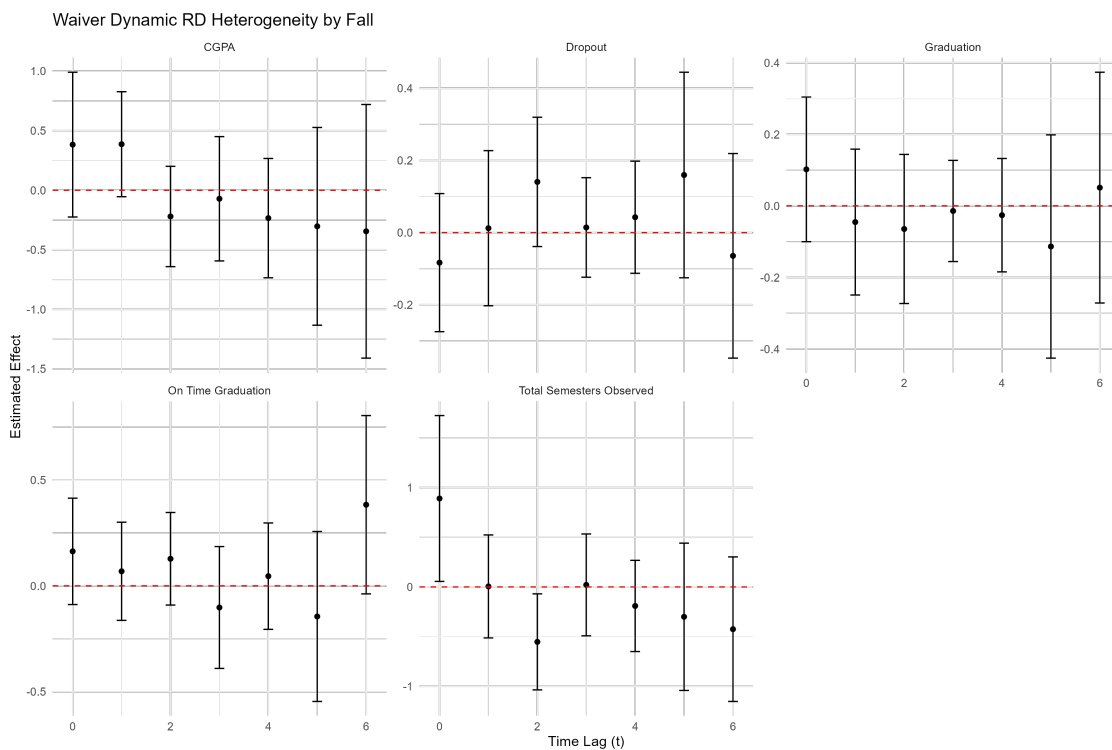
*Note:* These figures show the results of the one-shot estimator for waivers for undergraduate students with a program length of 8 semesters. The figures report the estimates on waiver receipt in past on current grades, with the x axis reporting the lag.

Figure 19: One-shot estimator results for stipends



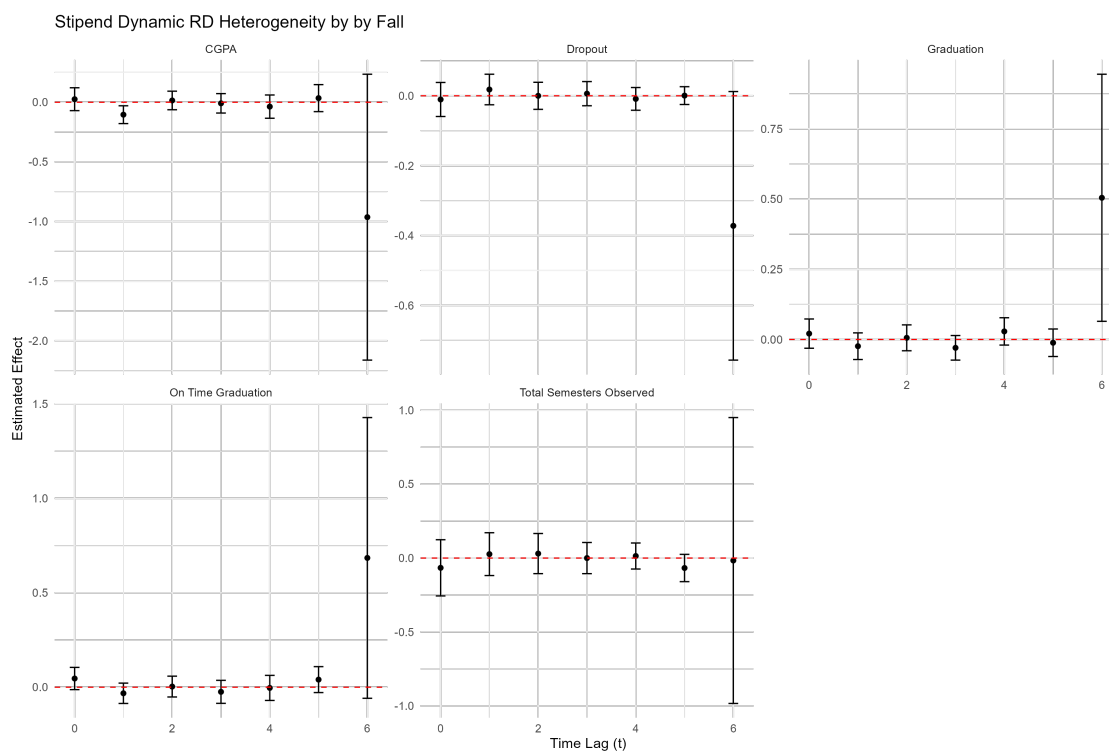
*Note:* These figures show the results of the one-shot estimator for stipends for undergraduate students with a program length of 8 semesters. The figures report the estimates on stipend receipt in past on current grades, with the x axis reporting the lag.

Figure 20: One-shot estimator heterogeneity for waivers in the fall



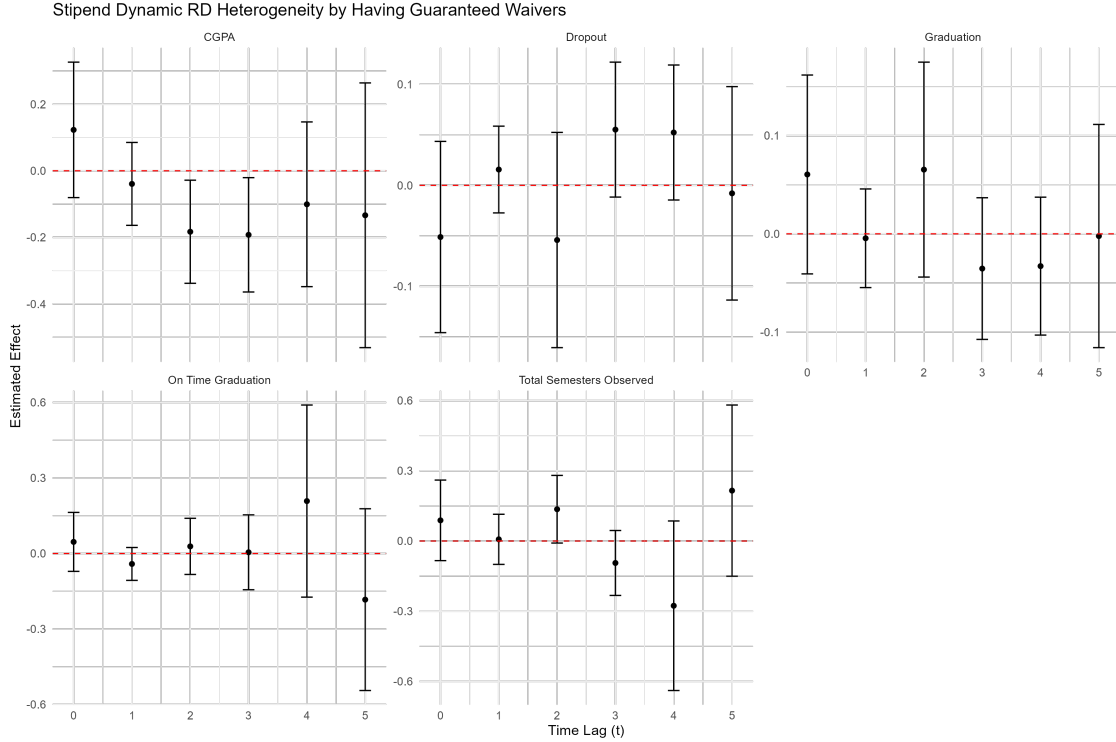
*Note:* These figures show the results of the interaction term between past aid and receiving it in the fall for the one-shot estimator for waivers for undergraduate students with a program length of 8 semesters. The figures report the estimates on waiver receipt in past on current grades, with the x axis reporting the lag.

Figure 21: One-shot estimator heterogeneity for stipends in the fall



*Note:* These figures show the results of the interaction term between past aid and receiving it in the fall for one-shot estimator for stipends for undergraduate students with a program length of 8 semesters. The figures report the estimates on stipend receipt in past on current grades, with the x axis reporting the lag.

Figure 22: One-shot estimator heterogeneity for stipends for those guaranteed aid



*Note:* These figures show the results of the interaction term between past aid and having a guaranteed waiver for one-shot estimator for stipends for undergraduate students with a program length of 8 semesters. The figures report the estimates on stipend receipt in past on current grades, with the x axis reporting the lag.

a guaranteed tuition waiver. This heterogeneity only applies to CGPA, with no significant difference for other variables such as dropout, graduation or persistence, implying that the main margin that stipends motivate students in RSU is through increased academic performance, but not affecting their academic progression.

## D Tournament Monte Carlo Simulation

To evaluate the potential biases in my dynamic RD estimates caused by violations of the assumption of additive separability, I conduct a Monte Carlo simulation. The data generating process simulates students competing for a fixed number of awards across repeated semesters.

Students have heterogeneous baseline ability and cumulative GPA following an AR(1)

process with an assumed persistence  $\rho = 0.70$  and an idiosyncratic shock  $\varepsilon_{it} \sim \mathcal{N}(0, 1 - \rho^2)$ . Winning an award delivers a direct GPA boost of  $\beta_1 = 0.25\sigma$ . The baseline semester-to-semester survival probability is 85%, and aid increases it by 5pp.

To estimate the bias caused by violations of additive separability, assume that current GPA is affected by past aid receipt:

$$GPA_{it} = \rho GPA_{i,t-1} + \beta_1 Aid_{it} + \gamma(Aid_{it} \times Aid_{i,t-1}) + \varepsilon_{it} \quad (12)$$

Where  $\gamma$  is the measure of how past aid receipt impacts the effect of aid receipt. A  $\gamma$  of 0 implies additive separability.

I simulate 150 tournaments for different values of  $\gamma$  from a range of -0.06 to 0.06, which are the maximum values I observe in Tables 7 and 8. For each of these scenarios, I estimate the total effect ( $\beta_2^T$ ), the renewal probability ( $\pi_{1,2}$ ), the marginal effect ( $\delta_2^M$ ), and compute the recursive estimator  $\hat{\delta}_{1,2}^{rec} \equiv \beta_2^T - \pi_{1,2}\delta_2^M$ . Table 21 reports the simulation results.

Table 21: Calibrated Tournament Monte Carlo Simulation: Robustness of the Recursive Estimator to Treatment-History Interactions

| Interaction Scenario             | Parameter ( $\gamma$ ) | Renewal Prob. ( $\pi_{1,2}$ ) | Total Effect ( $\beta_2^T$ ) | Marginal Effect ( $\delta_2^M$ ) | Recursive Est. ( $\hat{\delta}_{1,2}^{rec}$ ) | Shift vs. Baseline  |
|----------------------------------|------------------------|-------------------------------|------------------------------|----------------------------------|-----------------------------------------------|---------------------|
| Dynamic Substitutability         | -0.06                  | 0.676<br>(0.020)              | 1.970<br>(0.054)             | 2.165<br>(0.048)                 | 0.508<br>(0.038)                              | -2.9%               |
| Additive Separability (Baseline) | 0.00                   | 0.676<br>(0.021)              | 2.014<br>(0.059)             | 2.205<br>(0.051)                 | 0.523<br>(0.041)                              | Reference<br>(0.0%) |
| Dynamic Complementarity          | +0.06                  | 0.675<br>(0.020)              | 2.063<br>(0.054)             | 2.257<br>(0.047)                 | 0.539<br>(0.039)                              | +3.0%               |

Note: This table presents results from a calibrated Monte Carlo simulation of repeated financial aid tournaments across 150 replications per scenario (total  $N = 2,400$  students per replication across 80 risk sets of 30 students competing over 4 semesters). The data generating process is calibrated to the Latvian merit aid tournament data: AR(1) GPA persistence  $\rho = 0.70$ , baseline direct aid effect on GPA  $\beta_1 = 0.25\sigma$ , baseline survival probability 85%, and aid survival premium +5 percentage points. The parameter  $\gamma$  governs the interaction between current aid and lagged aid ( $Aid_{it} \times Aid_{i,t-1}$ ). The recursive decomposition estimator  $\hat{\delta}_{1,2}^{rec} = \beta_2^T - \pi_{1,2}\delta_2^M$  is evaluated at period 2. Monte Carlo standard deviations across replications are reported in parentheses. Across empirical ranges of  $\gamma \in [-0.06, +0.06]$  (matching the empirical bounds in Tables 7 and 8), the recursive estimator varies by only  $\pm 3.0\%$ , confirming that the dynamic decomposition is quantitatively robust to departures from additive separability.

What I find is that across the potential values of  $\gamma$ , the values of the estimator shift by 3% in either direction and do not materially change the conclusions, even if the assumption of additive separability is violated.

## E Institution-Specific Stipend Impacts (RSU and UL)

While the main text pools data across RSU and UL to maximize power, the institutional assignment mechanisms differ slightly. RSU incorporates scientific performance into its stipend assignments and guarantees waivers for enrolled students, while UL uses a strict GPA cutoff. This section disentangles the total impacts of stipend receipt by institution.

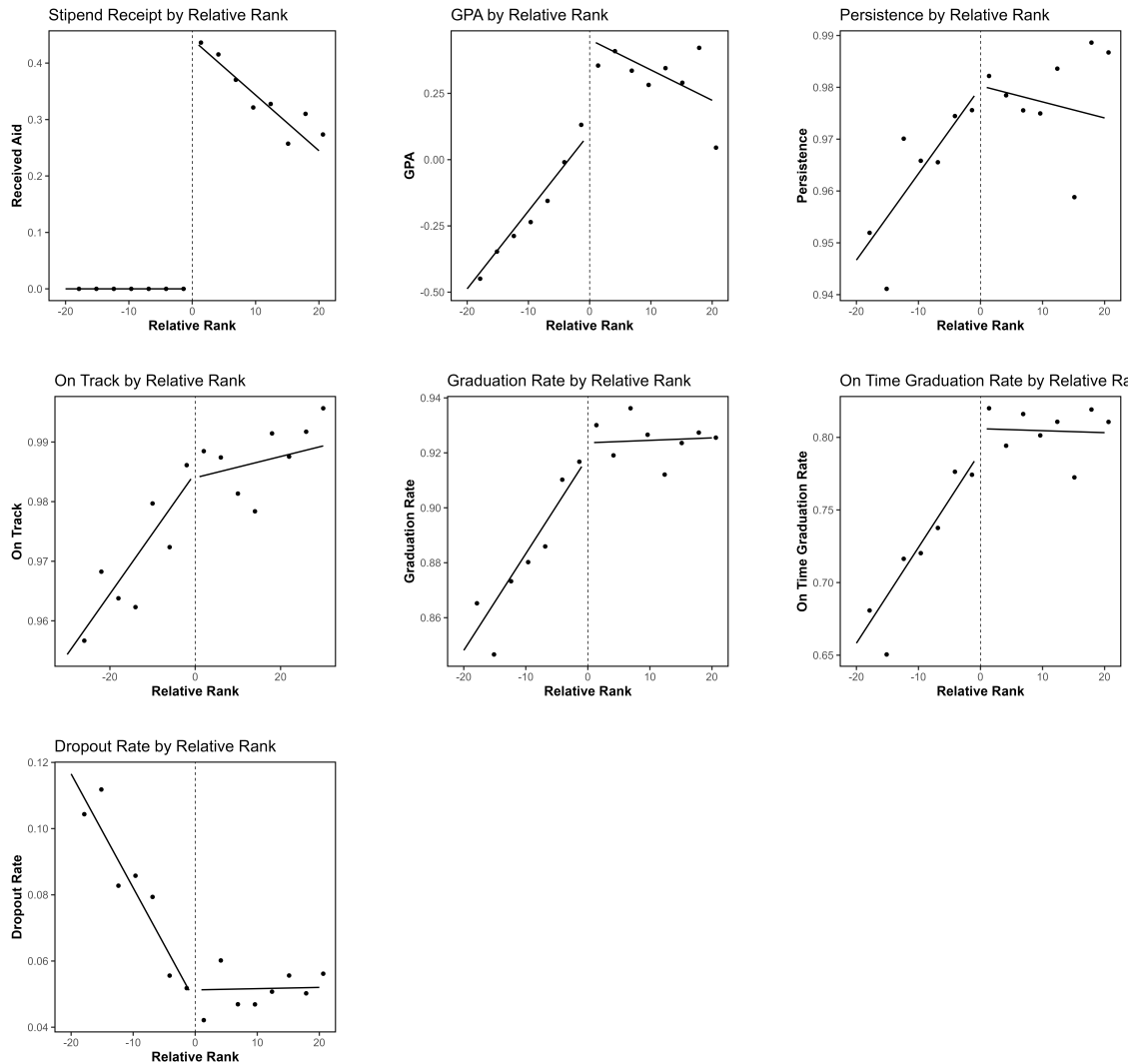
### E.1 Results for RSU

This subsection details the stipend impacts specifically for the RSU cohort. At RSU, a significant portion of the student body already holds a guaranteed tuition waiver upon enrollment. Consequently, the tournament for stipends operates under slightly different stakes than a purely open competition. As shown in Table 22, the overall impact of winning a stipend at RSU is concentrated primarily on the intensive margin, boosting standardized next-semester GPA by 0.306, with a weaker average effect on extensive margins like persistence or graduation.

However, exploring heterogeneity reveals nuanced dynamics for this specific institution. Table 24 demonstrates that for students in their first year, the impact of stipends on academic effort is positive (increasing GPA by  $0.114\sigma$ ), though it yields only marginal increases in persistence (1.5pp). The safety net of guaranteed waivers has behavioral effects at RSU. Table 25 shows that for students who already possess a guaranteed waiver, winning a stipend induces a complacency effect, decreasing next-semester GPA by 0.292 standard deviations with no significant effect on persistence. Interestingly, in more difficult risk sets (below-median GPA), winning a stipend induces students to significantly

improve their grades (by 0.128 standard deviations) but negatively impacts their persistence (-1.2 percentage points).

Figure 23: Student Outcomes Below and Above the Stipend Reception Threshold for RSU



*Note:* These figures show the change in student outcomes based on distance from the rank cutoff for reception of stipends. The plotted points are binned averages of the dependent variable, residualized by risk-set fixed effects. The lines are linear fits to the residualized data separately for each side of the cutoff.

Table 22: Effect of Stipends on Student Outcomes in the Next Semester

|                            | OLS                 | Risk Set FE         | 2SLS                | Per €1k             |
|----------------------------|---------------------|---------------------|---------------------|---------------------|
| GPA in Next Semester       | 0.175***<br>(0.028) | 0.175***<br>(0.028) | 0.306***<br>(0.048) | 0.569***<br>(0.090) |
| Persisted in Next Semester | 0.010*<br>(0.004)   | 0.002<br>(0.004)    | 0.004<br>(0.008)    | 0.007<br>(0.014)    |
| On Track in Next Semester  | 0.013***<br>(0.004) | 0.003<br>(0.004)    | 0.005<br>(0.007)    | 0.009<br>(0.013)    |
| Graduation                 | 0.055***<br>(0.012) | −0.004<br>(0.010)   | −0.006<br>(0.018)   | −0.012<br>(0.034)   |
| Dropped Out                | −0.024**<br>(0.008) | 0.011<br>(0.008)    | 0.020<br>(0.014)    | 0.037<br>(0.025)    |
| Graduation on Time         | 0.065***<br>(0.015) | 0.024+<br>(0.013)   | 0.042+<br>(0.024)   | 0.081+<br>(0.046)   |
| Risk Set FE                |                     | X                   | X                   | X                   |
| Num.Obs.                   | 16 360              | 16 360              | 16 360              | 16 360              |

+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

Note: The table reports intent-to-treat (reduced form) and instrumental variable (2SLS) estimates of the effect of stipend eligibility on student outcomes. 'Reduced Form' estimates are from a local linear regression of the outcome on the eligibility indicator. '2SLS' estimates instrument actual stipend receipt with eligibility. Persistence is defined as an indicator for whether the student is enrolled in the university in the following semester. On Track is defined as an indicator for whether the student progresses to the next academic semester in the following term (e.g. from semester 1 to semester 2). All models include risk set fixed effects and cluster standard errors at the student level. The minimum, maximum, and average first-stage F-statistics across the 2SLS models are 2627.4, 3222.6, and 2992.2 respectively.

Table 23: Heterogeneity by Student Demographics in the Effects of Stipends on Student Outcomes for RSU

| Outcome       | Heterogeneity Variable |                   |                      |                   |                   |
|---------------|------------------------|-------------------|----------------------|-------------------|-------------------|
|               | Fees                   | Male              | Age                  | Capital           | Minority          |
| GPA           | 0.000<br>(0.000)       | 0.068<br>(0.052)  | 0.012***<br>(0.003)  | −0.057<br>(0.035) | −0.046<br>(0.050) |
| Persistence   | 0.000<br>(0.000)       | −0.001<br>(0.007) | 0.000<br>(0.001)     | 0.006<br>(0.005)  | −0.008<br>(0.006) |
| On Track      | 0.000<br>(0.000)       | −0.004<br>(0.007) | 0.000<br>(0.001)     | 0.007<br>(0.005)  | −0.007<br>(0.006) |
| Dropout       | 0.000*<br>(0.000)      | 0.011<br>(0.015)  | −0.003***<br>(0.001) | 0.009<br>(0.009)  | −0.008<br>(0.014) |
| Graduation    | 0.000<br>(0.000)       | −0.025<br>(0.018) | 0.006***<br>(0.001)  | −0.011<br>(0.012) | −0.001<br>(0.017) |
| On-time Grad. | 0.000<br>(0.000)       | 0.004<br>(0.030)  | 0.005**<br>(0.002)   | −0.014<br>(0.018) | 0.006<br>(0.026)  |
| Risk Set FE   | X                      | X                 | X                    | X                 | X                 |
| Num.Obs.      | 15 534                 | 16 360            | 15 915               | 15 919            | 15 919            |

+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

Note: The table reports heterogeneity analysis of stipend eligibility effects. Each column reports the coefficient on the interaction term between the eligibility indicator and the column variable (e.g., 'Male'). Persistence is defined as an indicator for whether the student is enrolled in the university in the following semester. All regressions include risk set fixed effects, controls for the running variable, and the main effects of the heterogeneity variable. Robust standard errors clustered at the student level in parentheses.

Table 24: Heterogeneity by Cutoff Characteristics for RSU

| Outcome       | Heterogeneity Variable |                    |                     |                      |                    |
|---------------|------------------------|--------------------|---------------------|----------------------|--------------------|
|               | Cutoff Percentile      | First Year         | First Half          | Last Year            | Last Semester      |
| GPA           | 0.149+<br>(0.087)      | 0.114**<br>(0.035) | 0.090**<br>(0.033)  | 0.078*<br>(0.032)    | −0.041<br>(0.048)  |
| Persistence   | −0.057***<br>(0.014)   | 0.015**<br>(0.006) | 0.016**<br>(0.005)  | −0.018***<br>(0.005) | 0.005<br>(0.007)   |
| On Track      | −0.055***<br>(0.013)   | 0.016**<br>(0.005) | 0.016***<br>(0.005) | −0.017***<br>(0.004) | 0.004<br>(0.007)   |
| Dropout       | 0.103***<br>(0.023)    | −0.019<br>(0.013)  | −0.014<br>(0.009)   | 0.014+<br>(0.009)    | −0.019+<br>(0.010) |
| Graduation    | −0.086**<br>(0.029)    | 0.026<br>(0.017)   | 0.022+<br>(0.012)   | −0.018<br>(0.011)    | 0.012<br>(0.012)   |
| On-time Grad. | −0.112**<br>(0.041)    | 0.014<br>(0.020)   | 0.031*<br>(0.015)   | −0.029+<br>(0.015)   | −0.001<br>(0.018)  |
| Risk Set FE   | X                      | X                  | X                   | X                    | X                  |
| Num.Obs.      | 16 360                 | 16 360             | 16 327              | 16 327               | 16 327             |

+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

Note: The table reports heterogeneity analysis by characteristics of the cutoff itself. Each column reports the coefficient on the interaction term between eligibility and the characteristic. 'Cutoff Percentile' is the percentile rank of the cutoff within the risk set. 'First Year' indicates if the student is in their first year of study. Persistence is defined as an indicator for whether the student is enrolled in the university in the following semester. Robust standard errors clustered at the student level in parentheses.

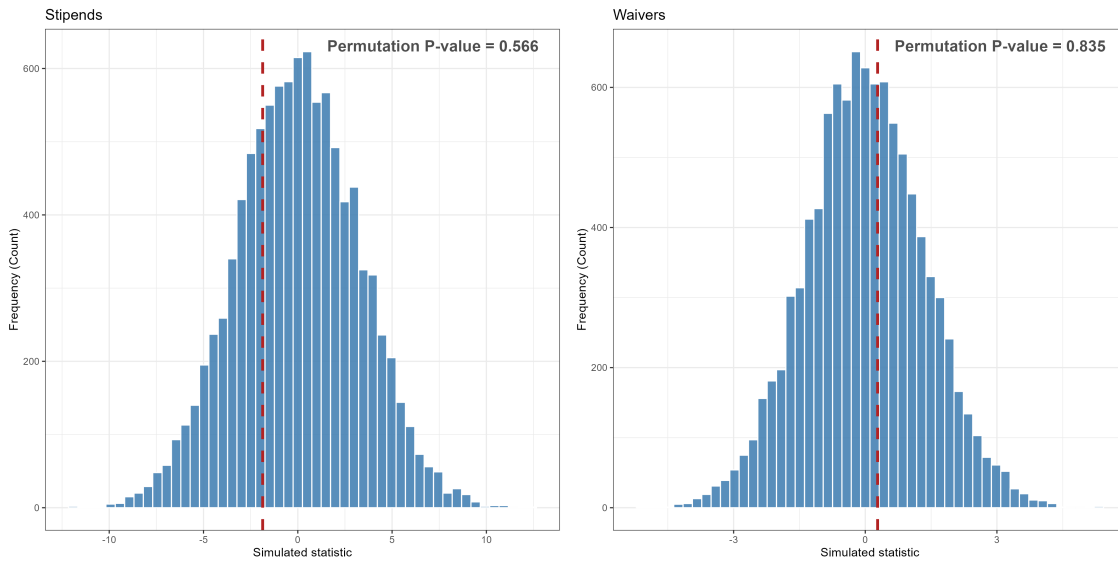
Table 25: Heterogeneity by Competitiveness in the Effects of Stipends on Student Outcomes for RSU

| Outcome       | Heterogeneity Variable |                     |                     |                    |
|---------------|------------------------|---------------------|---------------------|--------------------|
|               | Fall                   | <Median GPA         | <Median Progression | Guaranteed Aid     |
| GPA           | −0.030<br>(0.053)      | 0.128*<br>(0.055)   | −0.015<br>(0.056)   | −0.292*<br>(0.129) |
| Persistence   | −0.012<br>(0.008)      | −0.015+<br>(0.008)  | 0.008<br>(0.008)    | 0.004<br>(0.014)   |
| On Track      | −0.012<br>(0.008)      | −0.021**<br>(0.008) | 0.014+<br>(0.008)   | 0.003<br>(0.014)   |
| Dropout       | −0.003<br>(0.014)      | 0.007<br>(0.015)    | −0.013<br>(0.014)   | −0.023<br>(0.037)  |
| Graduation    | 0.019<br>(0.017)       | −0.031<br>(0.020)   | 0.001<br>(0.018)    | −0.002<br>(0.057)  |
| On-time Grad. | 0.027<br>(0.024)       | −0.048+<br>(0.027)  | 0.017<br>(0.025)    | 0.004<br>(0.073)   |
| Risk Set FE   | X                      | X                   | X                   | X                  |
| Num.Obs.      | 16 360                 | 16 360              | 16 360              | 16 360             |

+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

Note: The table reports heterogeneity analysis by various measures of competitiveness specifically for RSU. Each column reports the coefficient on the interaction term between aid eligibility and the column variable (*Above*  $\times$  *Het*). All models include risk set fixed effects and allow separate linear trends by subgroup on both sides of the cut-off (*Above*  $\times$  *Het*  $\times$  *Runvar*). 'Fall' indicates the Fall semester. '<Median GPA' indicates below-median baseline GPA. '<Median Progression' indicates higher difficulty risk sets based on a predetermined, leave-one-cohort-out historical on-track rate for the same study program and semester. 'Guaranteed Aid' indicates students who started with a waiver. Robust standard errors clustered at the student level in parentheses.

Figure 24: Distribution of Density Test Values for RSU



*Note:* These figures show the distribution of density test values for the dataset wherein the cutoff values are randomly reassigned within risk sets. Line indicates density test value for observed data.

## E.2 Results for UL

This subsection details the stipend impacts specifically for the UL cohort. Unlike RSU, UL utilizes a strict, GPA-only cutoff for stipend eligibility and does not offer the same widespread safety net of guaranteed waivers. This creates a sharper tournament environment where the stakes for academic performance are higher.

Consequently, the baseline 2SLS estimates for UL (Table 27) reveal impacts across both the intensive margin (GPA increases by  $0.213 \sigma$ ) and the extensive margins (persistence increases by 6.9pp, and graduation by 9.3pp). For these students, the liquidity provided by a stipend plays a critical role in ensuring they can remain enrolled and complete their degrees.

Heterogeneity analysis further underscores the retention value of stipends in high-stakes environments. Table 28 shows that in less competitive or struggling cohorts (below-median GPA), stipends have smaller impacts on student GPA and graduation, but below-median progression risk sets see larger impacts on persistence. Similarly to RSU, the timing of the award matters (Table 29): early stipends in the first half of a student's studies

Table 26: Balance Test for UL

|             | No FE              | Risk Set FE        |
|-------------|--------------------|--------------------|
| Female      | −0.020+<br>(0.011) | −0.014<br>(0.010)  |
| Age         | 0.145<br>(0.115)   | 0.288**<br>(0.096) |
| Capital     | 0.011<br>(0.013)   | 0.016<br>(0.013)   |
| Minority    | 0.001<br>(0.009)   | −0.001<br>(0.008)  |
| Math CE     | 2.049**<br>(0.718) | 0.883<br>(0.569)   |
| Risk Set FE |                    | X                  |
| Num.Obs.    | 23 574             | 23 574             |

+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

Note: The table reports balance tests testing for discontinuities in baseline characteristics at the cutoff. Estimates are from a local linear regression with a uniform kernel within the optimal bandwidth. The running variable is the student's relative rank. 'Risk Set FE' indicates the inclusion of risk set fixed effects. Standard errors are clustered at the student level.

increase persistence, while late stipends primarily drive final GPA.

Figure 25: Distribution of Density Test Values for UL

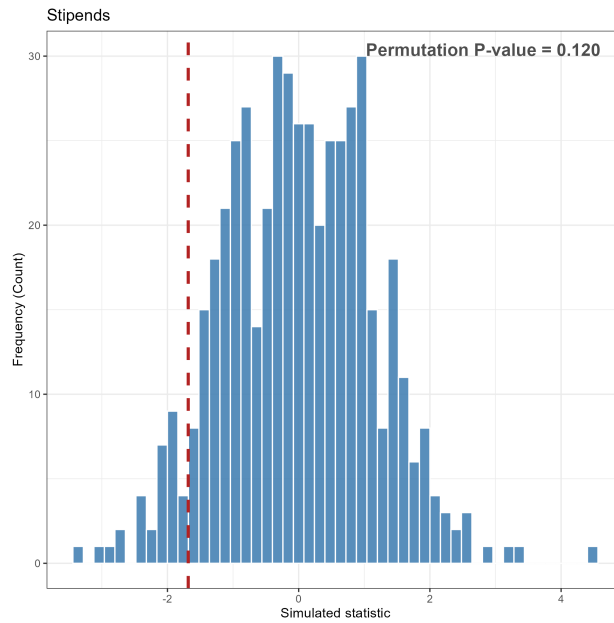
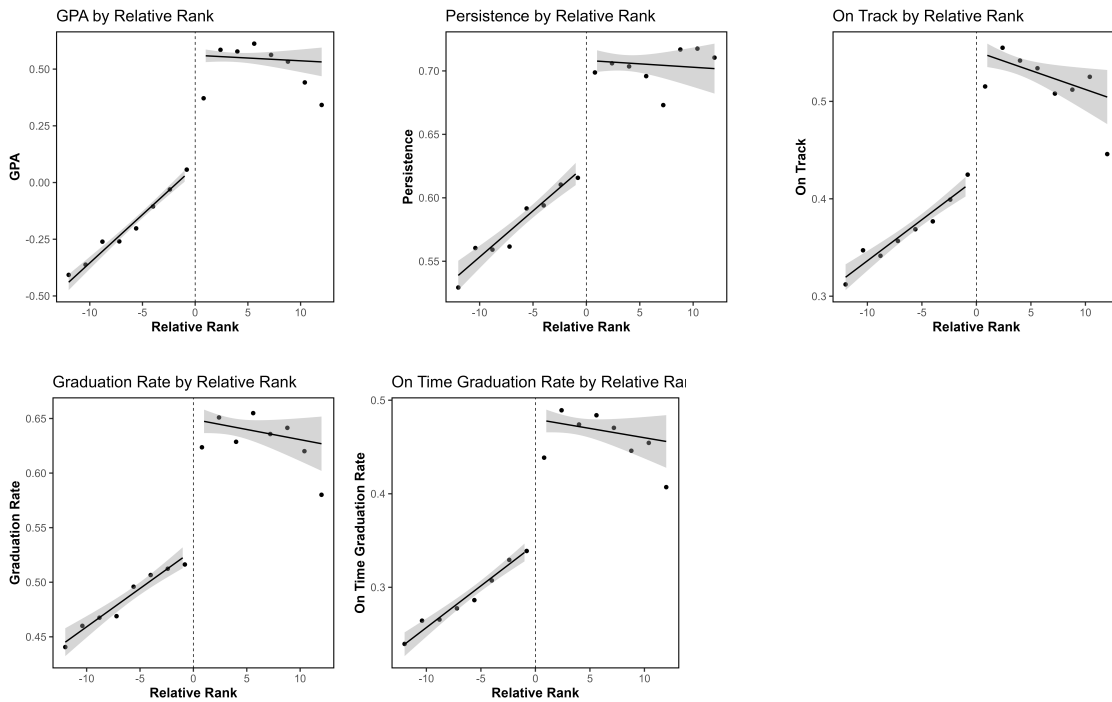


Figure 26: Student Outcomes Below and Above the Stipend Reception Threshold for UL



*Note:* These figures show the change in student outcomes based on distance from the rank cutoff for the reception of stipends. The plotted points are binned averages of the dependent variable, residualized by risk-set fixed effects. The lines are linear fits to the residualized data separately for each side of the cutoff.

Table 27: Effect of Stipends on Student Outcomes in the Next Semester in UL

|                            | OLS                 | Risk Set FE         | 2SLS                | Per €1k             |
|----------------------------|---------------------|---------------------|---------------------|---------------------|
| GPA in Next Semester       | 0.253***<br>(0.026) | 0.204***<br>(0.025) | 0.213***<br>(0.026) | 0.428***<br>(0.053) |
| Persisted in Next Semester | 0.066***<br>(0.011) | 0.066***<br>(0.008) | 0.069***<br>(0.009) | 0.135***<br>(0.017) |
| On Track in Next Semester  | 0.083***<br>(0.012) | 0.076***<br>(0.011) | 0.080***<br>(0.012) | 0.182***<br>(0.021) |
| Graduation                 | 0.116***<br>(0.012) | 0.088***<br>(0.010) | 0.093***<br>(0.011) | 0.156***<br>(0.023) |
| Graduation on Time         | 0.091***<br>(0.012) | 0.070***<br>(0.011) | 0.074***<br>(0.012) | 0.145***<br>(0.023) |
| Risk Set FE                |                     | X                   | X                   | X                   |
| Num.Obs.                   | 24 231              | 24 231              | 24 231              | 24 231              |

+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

Note: The table reports estimates for various student outcomes. 'OLS' is the simple reduced-form estimate. 'Risk Set FE' includes fixed effects. '2SLS' instruments aid receipt with eligibility at the GPA cutoff. The minimum, maximum, and average first-stage F-statistics across the 2SLS models are 52894.9, 65129.5, and 61032.3 respectively.

Table 28: Heterogeneity by Competitiveness in the Effects of Stipends on Student Outcomes in UL

|             | <Median GPA         | <Median Progression |
|-------------|---------------------|---------------------|
| GPA         | −0.141**<br>(0.050) | 0.053<br>(0.051)    |
| Persistence | 0.019<br>(0.018)    | 0.032+<br>(0.017)   |
| On Track    | −0.016<br>(0.023)   | −0.005<br>(0.022)   |
| Graduation  | −0.036+<br>(0.021)  | 0.007<br>(0.020)    |
| On time     | −0.016<br>(0.022)   | −0.012<br>(0.021)   |
| Aid Receipt | −0.017*<br>(0.007)  | −0.002<br>(0.007)   |
| Risk Set FE | X                   | X                   |
| Num.Obs.    | 24 231              | 24 231              |

+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

Note: The table reports heterogeneity analysis of aid eligibility effects by measures of competitiveness. Each column reports the coefficient on the interaction term between the eligibility indicator and the column variable. High difficulty is defined as a risk set being below the median raw GPA or rate of students progressing on track. All regressions include risk set fixed effects, controls for the running variable, and the main effects of the heterogeneity variable. Robust standard errors clustered at the student level in parentheses.

Table 29: Heterogeneity by Cutoff Characteristics for UL

| Outcome     | Heterogeneity Variable |                      |                     |                      |
|-------------|------------------------|----------------------|---------------------|----------------------|
|             | Cutoff Percentile      | First Half           | Last Year           | Last Semester        |
| GPA         | −0.064<br>(0.150)      | −0.187***<br>(0.050) | 0.083<br>(0.054)    | 0.791<br>(0.586)     |
| Persistence | −0.016<br>(0.054)      | 0.045**<br>(0.016)   | −0.042**<br>(0.015) | −0.071***<br>(0.011) |
| On Track    | −0.052<br>(0.062)      | 0.028<br>(0.022)     | −0.017<br>(0.023)   | −0.090**<br>(0.031)  |
| Graduation  | −0.044<br>(0.059)      | 0.016<br>(0.020)     | −0.013<br>(0.019)   | −0.085***<br>(0.017) |
| Ontime      | −0.026<br>(0.057)      | 0.030<br>(0.021)     | −0.019<br>(0.022)   | −0.083**<br>(0.030)  |
| Aid Receipt | 0.400***<br>(0.035)    | 0.000<br>(0.007)     | 0.010<br>(0.007)    | 0.002<br>(0.010)     |
| Risk Set FE | X                      | X                    | X                   | X                    |
| Num.Obs.    | 24 231                 | 24 231               | 24 231              | 24 231               |

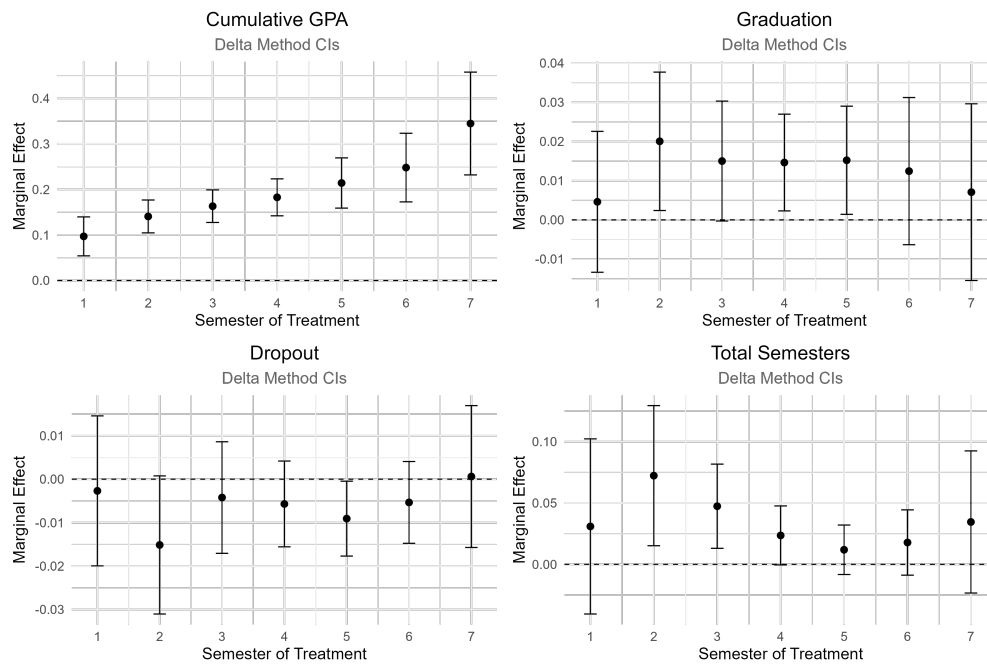
+  $p < 0.1$ , \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$

Note: The table reports the estimates from a series of regressions estimating the impact of being eligible financial aid on student grades, persistence, being on track in the next semester, graduation and graduation on time. Each estimate is from a separate regression. Persistence is defined as the student still being enrolled in a given semester. Robust standard errors clustered at the student level in parentheses.

## F Dynamic Decomposition Results for Stipends in RSU

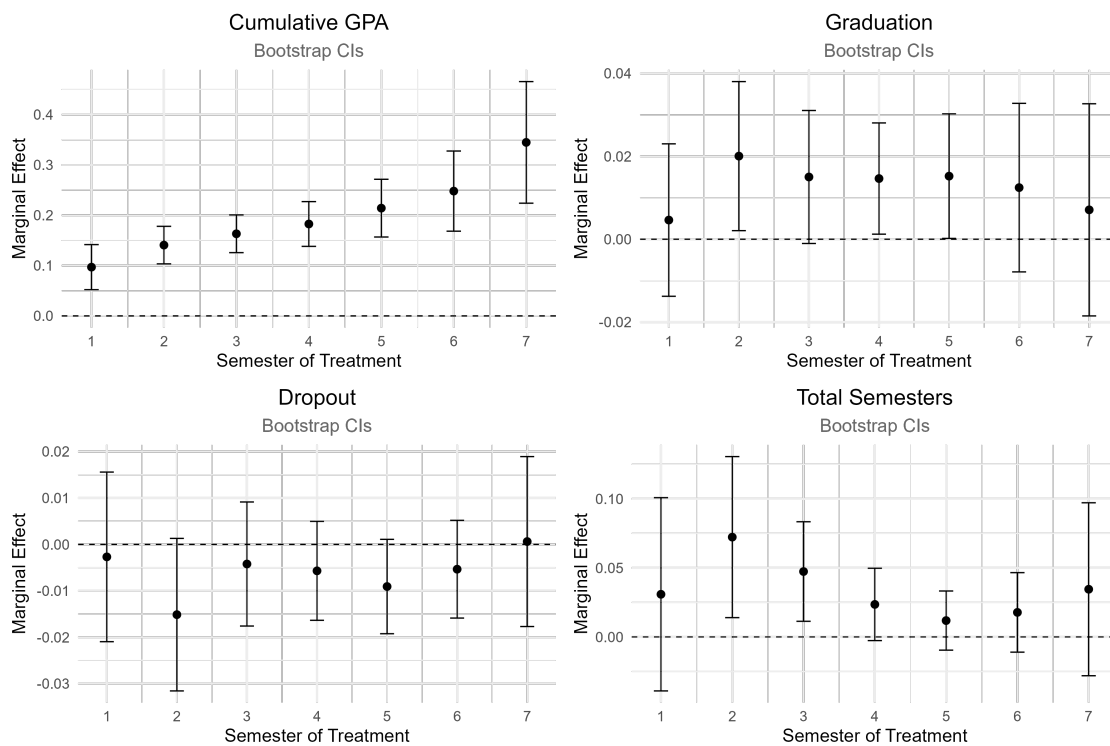
This appendix reports the dynamic regression discontinuity decomposition results for cash stipends specifically among students enrolled at RSU. Figure 27 displays marginal effect estimates with Delta method standard errors, Figure 28 reports bootstrapped confidence intervals, and Figure 29 displays the full decomposition into direct marginal and indirect crowd-in effects.

Figure 27: Marginal Effect of Stipend Eligibility at a Given Semester with Delta Method SEs



*Note:* The figures report the estimates from an indirect regression discontinuity decomposition on the marginal impact of being eligible for stipends in any given semester on a range of student outcomes. The estimation is done on students who start during the time period and whose expected graduation falls before the data horizon, and whose program length is 8 semesters.

Figure 28: Marginal Effect of Stipend Eligibility at a Given Semester with Bootstrapped SEs



*Note:* The figures report the estimates from an indirect regression discontinuity decomposition on the marginal, indirect, and total impact of being eligible for stipends in any given semester on a range of student outcomes. The estimation is done on students who start during the time period and whose expected graduation falls before the data horizon, and whose program length is 8 semesters.

Figure 29: Decomposition of Impact of Stipend Eligibility



*Note:* The figures report the estimates from an indirect regression discontinuity decomposition on the marginal, indirect, and total impact of being eligible for stipends in any given semester on a range of student outcomes. The estimation is done on students who start during the time period and whose expected graduation falls before the data horizon, and whose program length is 8 semesters.